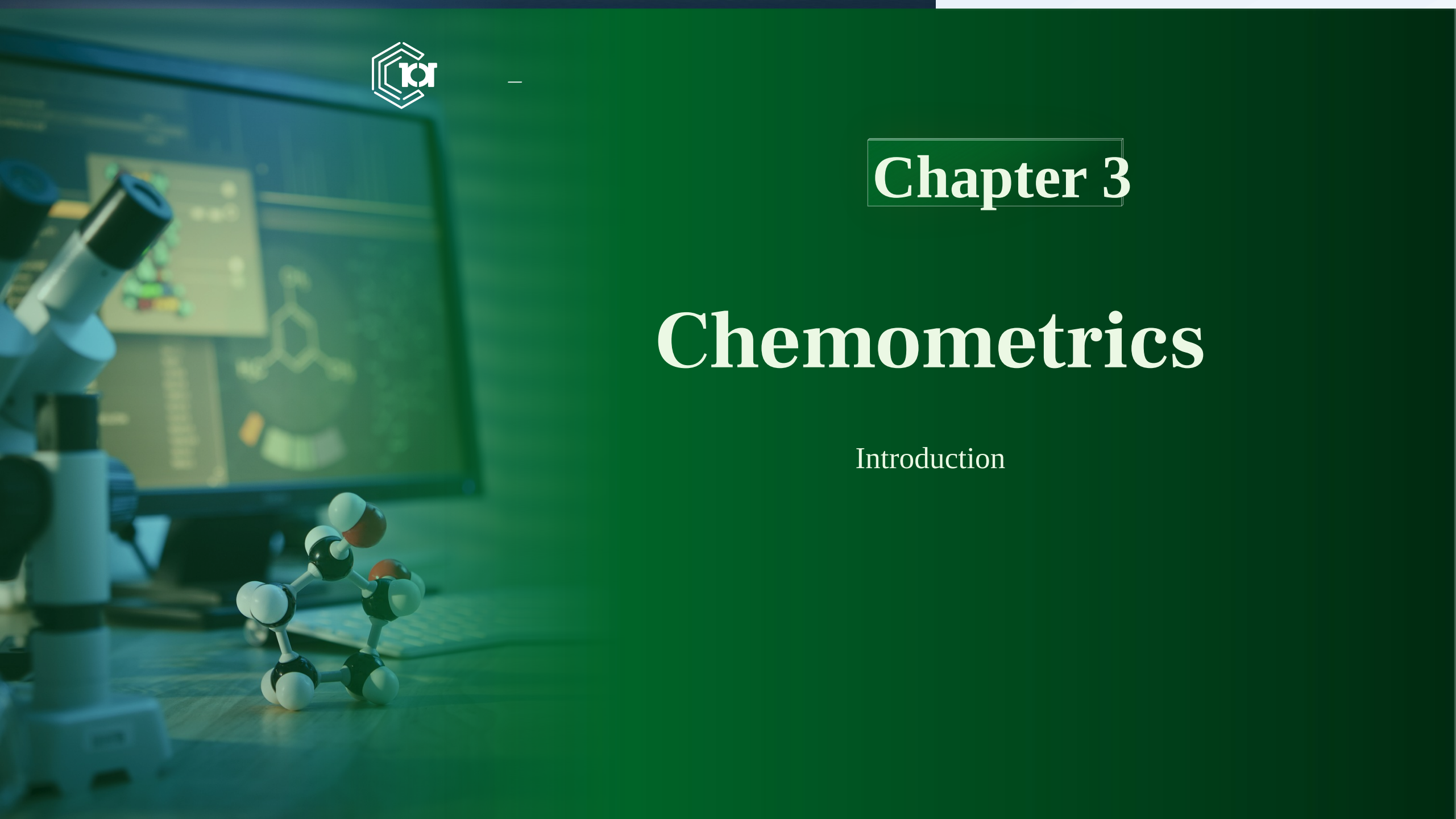




Chapter 3

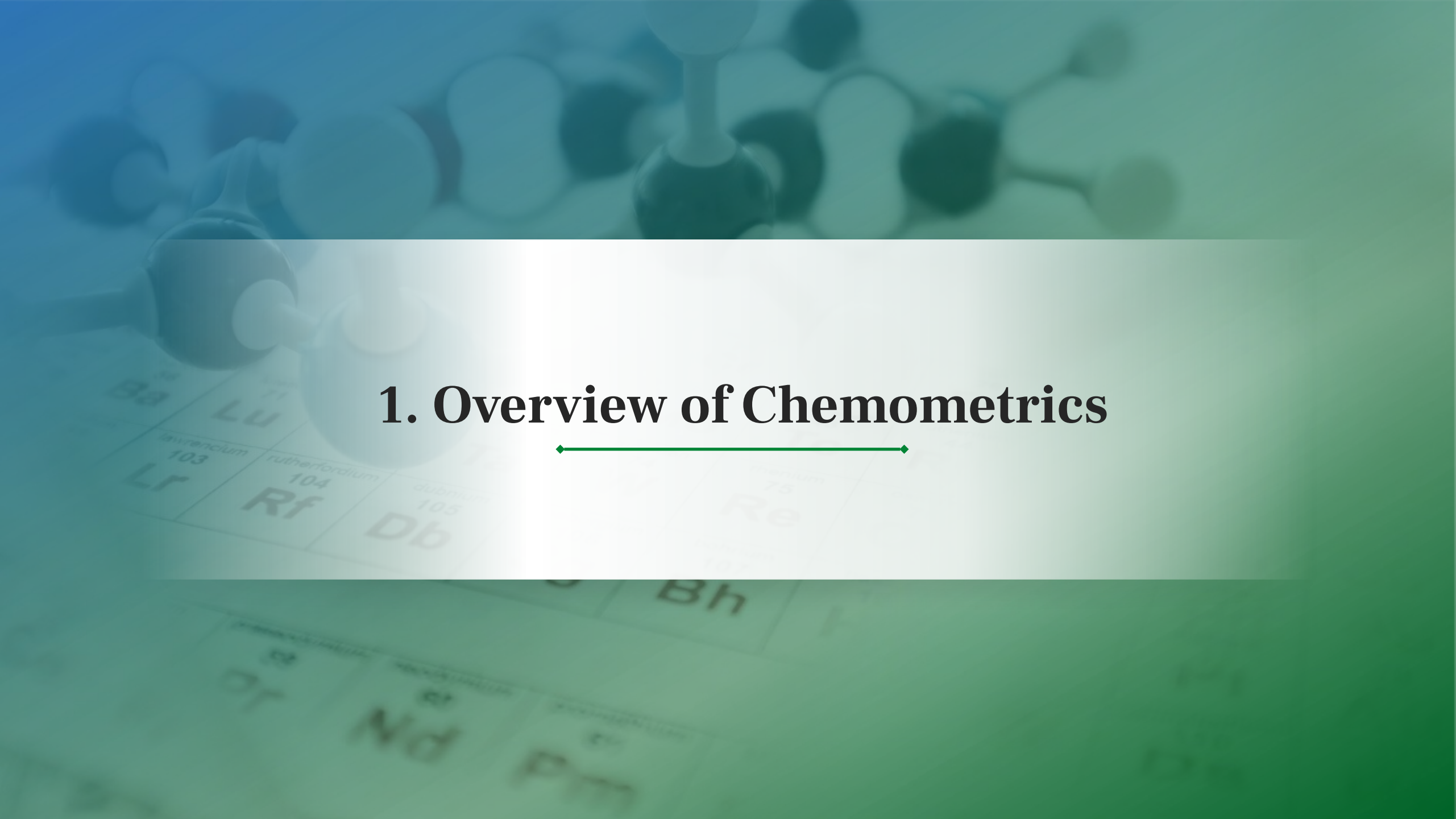
Chemometrics

Introduction



Contents

- 1. Overview of Chemometrics**
- 2. Spectroscopy Combined with Chemometrics**
- 3. Software and codes**



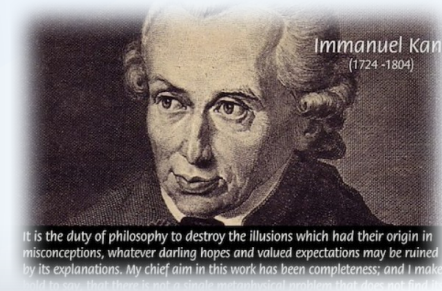
1. Overview of Chemometrics

Why Chemometrics ()?



“Among the branches of natural sciences, only those that can be expressed in mathematics are true sciences”.

—Immanuel Kant



Evolutions in Analytical Chemistry:

1. Equilibrium chemistry:
2. Computer: Process controlling, Chemometrics, Hyphenated instrument
3. Interdisciplinarity: Biology, environment, food....

Origin, Definition, and Development History



Chemometrics was born in the early 1970s. It is usually defined as “Chemometrics is a branch of chemistry, which uses *mathematical and statistical methods* with computer technology, designs and selects the best measurement procedures and experimental methods, in order to **obtain the maximum information by interpreting chemical data**”.

Chemometrics

Computer in Chemistry

Chemical Data Analysis



Svante WOLD



Bruce R. Kowalski

<https://doi.org/10.1002/cem.2622>

Origin, Definition, and Development History



➤ *The early chemometrics were mostly from the classical statistical methods.*

For example, the concept of **principal component analysis (PCA)** was proposed by British statistician K. Pearson early as in 1901, and was later developed and popularized by American statistician H. Hotelling in 1933. Till 1972, PCA was then used for deconvolution of chromatographic overlapping peaks. The famous **partial least squares (PLS)** was proposed by H. Wold, an econometric statistician in Sweden, for processing economic data in the 1960s. Later, his son S. Wold developed it in 1983 to solve the difficult chemical data regression problem, and obtained very satisfactory results....

➤ *The flourishing period of chemometrics was in the 1980s.*

The popularity of computers, drive of **industrial interests** (pharmaceutical development and process analysis), upgrade of **analytical instruments**, together made the research of chemometrics reach an unprecedented depth and breadth. In fact, some of the foremost methods now widely used are mostly created or perfected at that time.

Origin, Definition, and Development History



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Development of chemometrics can be roughly divided into four stage:

➤ *Pre-establishment.*

The characteristic of this stage is the application of mathematical statistics in chemistry, especially analytical chemistry. Analysts discussed the standard deviation, confidence interval, least square regression, and other issues of the analysis results. Organic chemists studied the structure-activity relationship of linear free energy, which can be considered the predecessor of chemical quantitative structure-activity relationship (QSAR). In general, the mathematical methods used by analysts during this period are basically descriptive. However, in other disciplines such as engineering science, psychology and other behavioral sciences, factor analysis, pattern recognition, and other methods have been used for higher-level data processing.

➤ *Birth of chemometrics.*

Chemometrics had become a major branch of analytical chemistry. This development includes two factors. One is the gradual popularization of computers, including the instrumentation of analytical chemistry that can accurately provide chemists with a large amount of reliable data. How to efficiently convert the data of these instruments into useful information naturally became the original drive for developing chemometrics. The second one is that various powerful mathematical methods can be applied in analytical chemistry with the help of faster computing.

Origin, Definition, and Development History



➤ *1980s: The mature of different theories, algorithm, and software.*

The unique multivariate calibration, multivariate discrimination, and chemical pattern recognition methods, such as partial least squares, soft independent modeling of class analogy (SIMCA), rank annihilation factor analysis, evolving factor analysis, etc., had been greatly developed in theory and algorithm research. In 1984,

American Mathwork Company officially launched MATLAB software, by which many complex mathematical calculations used in chemometrics can be realized with only one coding expression, making it almost a standard programming language for chemometrics research.

➤ *1990s: Towards practical applications*

Chemometrics had truly entered the stage of practical applications, such as near-infrared spectroscopy, sensors, medicine and pharmacy, etc. Almost all modern analytical instruments had a computer or microprocessor containing the chemometrics software. Chemometrics was becoming an indispensable tool in the daily work of chemistry or analytical chemistry. Nowadays,

Thinking: Why we need to introduce AI in to chemometrics...

Content of Chemometrics



Its content almost covers the entire process of chemical measurement.

Measurement

Sampling

Experiment

Measurement

Identification
quantification

Extraction

Decision

Sampling
Theory

Experiment
design

Process and
analyze signal

Multivariate
calibration
Pattern recognition

QSAR
Computer
simulation

AI

Chemometrics

Content of Chemometrics



Its content almost covers the entire process of chemical measurement.

Measurement

Sampling

Experiment

Measurement

Identification
quantification

Extraction

Decision

Where is math

Sampling
Theory

Experiment
design

Process and
analyze signal

Multivariate
calibration
Pattern recognition

QSAR
Computer
simulation

AI

Chemometrics

Content of Chemometrics



1. Sampling theory and method

Sampling is the first step of analysis. The reliability of analytical results is directly related to whether the sampling is correct or rational. The purpose of analysis or testing is to obtain the unbiased information about the entire object based on the data measured from a sectional sample.

2. Experimental design and optimization

Experimental design and optimization need to design and arrange experiments and optimize measurement conditions so as to improve work efficiency. Orthogonal design and simplex optimization method are still the main strategy for experimental optimization. Its purpose is to obtain as much information as possible about the relationship between the target and the factors with the fewest number of trials.

3. Signal processing

Interference signal and noise are often mixed in the analysis signal. By use of signal smoothing, filtering, transformation, peak splitting, curve fitting, derivation, and integration techniques, analysis signals can be reliably distinguished and detected from interference signals, and the signal-to-noise ratio can be improved.

4. Resolution and calibration

Multivariate resolution and calibration are the core content and also the most distinctive part of chemometrics. Calibration is a mathematical process that extracts useful information from the instrument signal. Its purpose is to establish the relationship between the analysis signal and the concentration for the quantification of the analyte. Multivariate calibration is a method used to improve the selectivity and reliability of analysis that is suitable for a variety of instrument signals, such as spectrum, mass spectrum, and chromatographic data. It correlates the independent variable (measurement information) of the training set with the dependent variable (the property of interest, such as the concentration of an analyte in a complex system or other physical and chemical properties) so as to establish multivariate calibration models.

5. Pattern recognition

Chemical pattern recognition is to select the characteristics of samples, find the rules of classification, and then classify and identify unknown sample sets according to the rules of classification. If sample is known, then classified; if unknown, the classification depends entirely on the natural characteristics of the sample.

6. Computer simulation

Simulation is an important means of using computer to study chemical reactions, measuring methods, and analyzing data. Monte Carlo simulation is one of the most commonly used simulation methods.

7. Quantitative structure-activity relationships

Quantitative structure-activity relationship (QSAR) uses multivariate calibration and pattern recognition methods to find out the quantitative relationship between structure, properties, and biological activity from a series of compounds with the already known activities, then predict the activity of new compounds, and guide the design of new compounds.

8. Chemical database and library searching

With the daily increase of spectrum data, various databases appeared, such as compound structure databases, various spectrum databases, physical property databases, etc. The rapid retrieval and effective use of data have become an important research content of computer processing information.

9. Artificial intelligence and chemical expert system

The chemical expert system is an intelligent computer program system that applies chemical knowledge and logical reasoning to solve chemical problems. It covers molecular structure analysis, selection of the best measurement, and separation conditions for various instruments (chromatography, spectroscopy, etc.), etc.

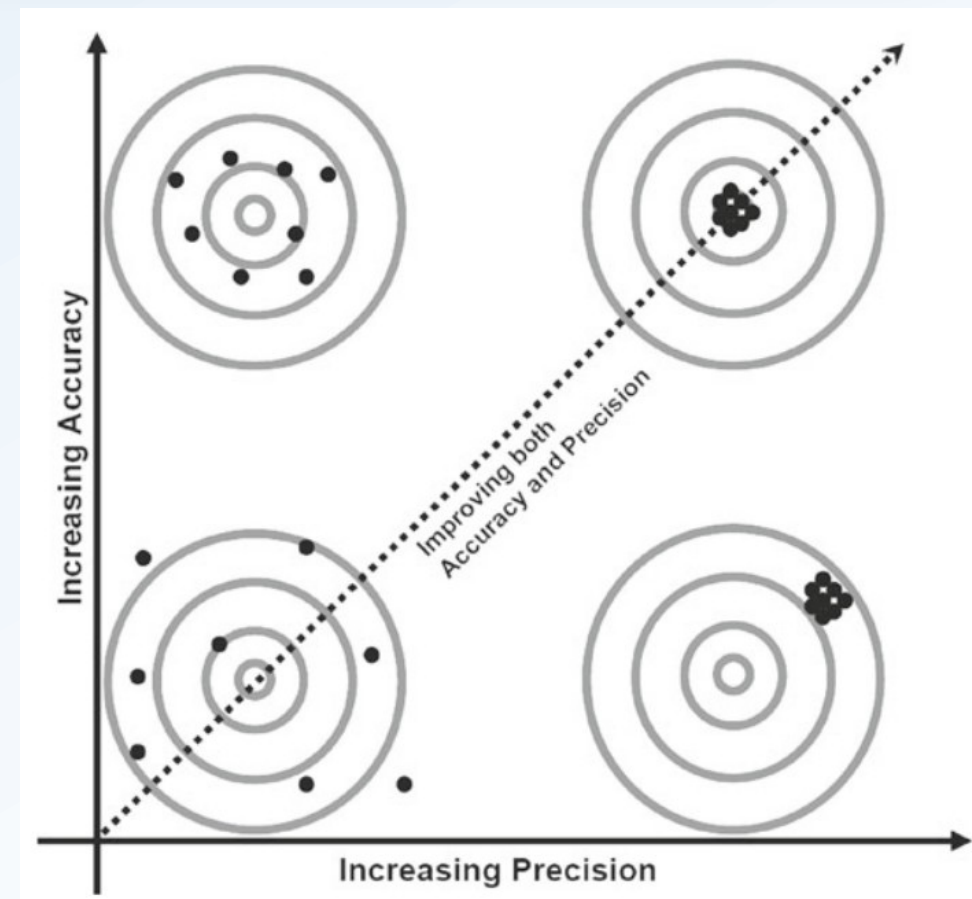
Necessity of Chemometrics



Application of chemometrics to the quantitative and qualitative analyses of spectroscopy in many cases makes the analysis result a significant level-up.

Necessary!

Multivariate calibration (), can improve the accuracy and precision of analysis. Factor analysis methods such as principal component regression and partial least squares can not only make use of the full spectrum but also significantly reduce the interference of coexisting components and background. The concentration of multiple components can be directly determined without chemical separation.



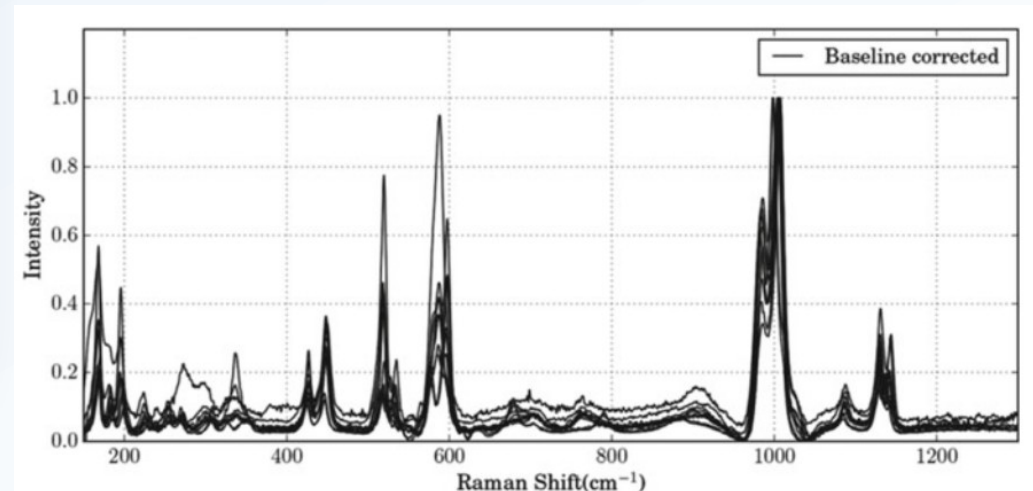
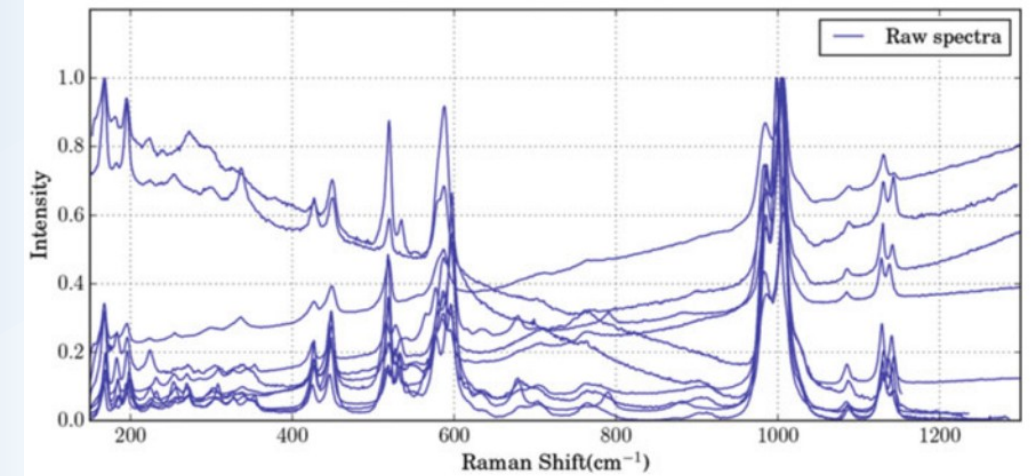
Necessity of Chemometrics



Signal processing technology

can improve the S/N ratio of the instrument, increase sensitivity, eliminate interference, extract useful information hidden in the spectrum, separate overlapping peaks, and improve resolution of the spectrum. For example, methods as Fourier and wavelet transform can smooth, denoise, and compress the spectrum, reliably distinguish and detect useful signals from the interferences, providing high-quality characteristic variables for multivariate calibration.

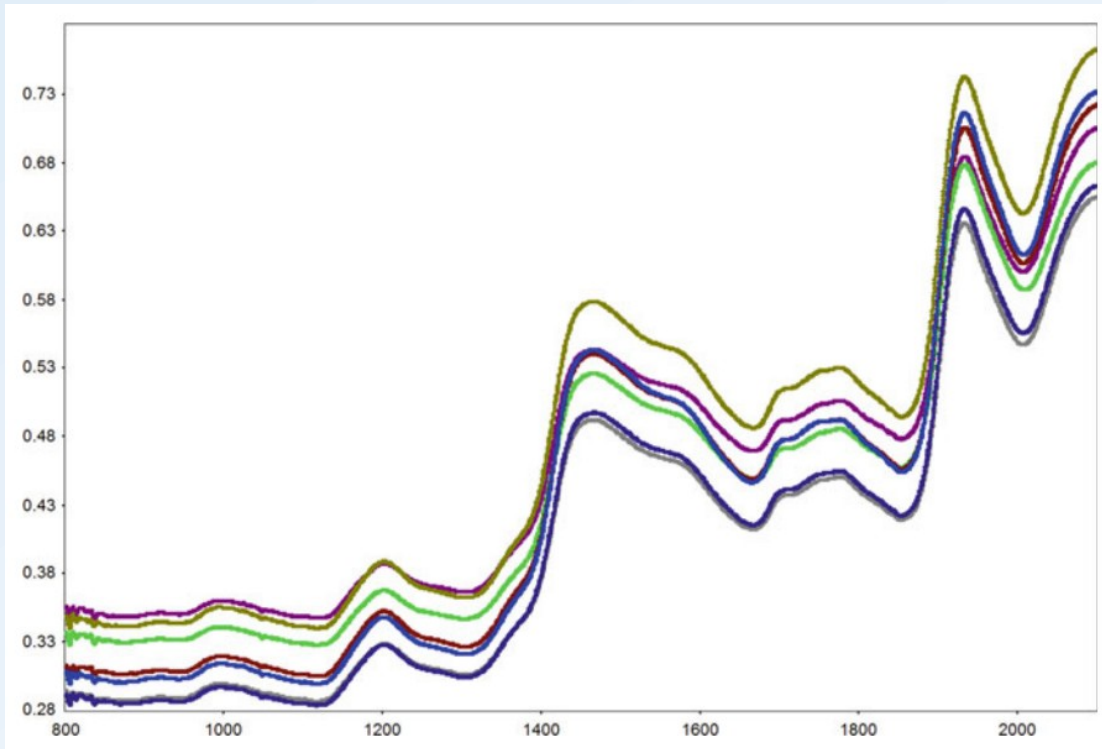
- Ten original Raman spectra of the same mineral from different origins in the **RRUFF** mineral database



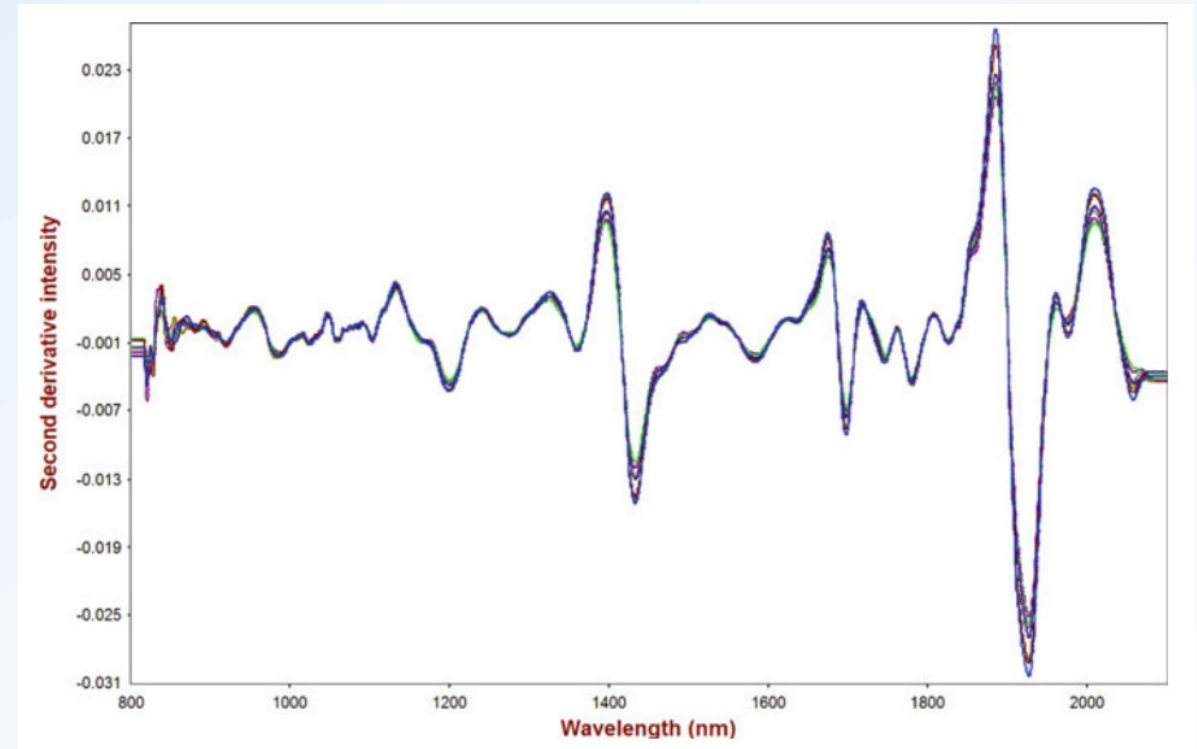
Necessity of Chemometrics



NIR spectra of flour



Second derivative processing



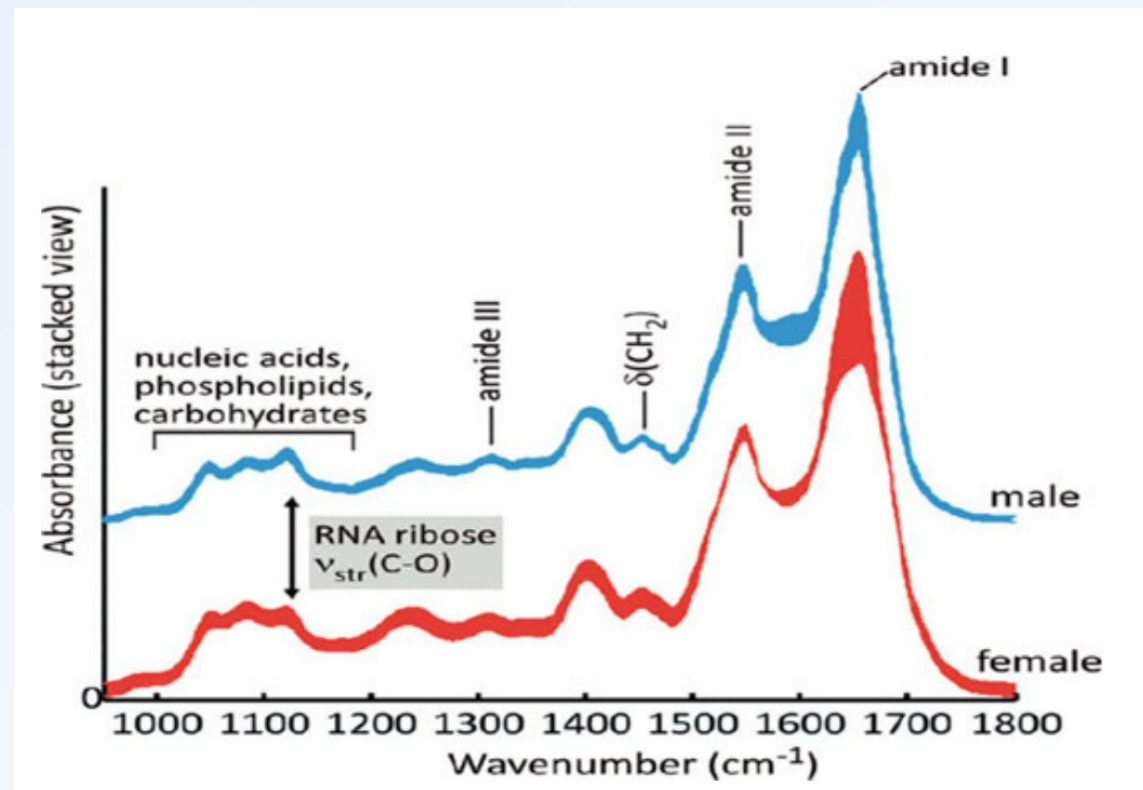
- It can be seen that this method effectively enhances the reflection characteristics of the spectral curve and provides the possibility for further analysis of the chemical composition of the lunar mantle!

Necessity of Chemometrics



Pattern recognition

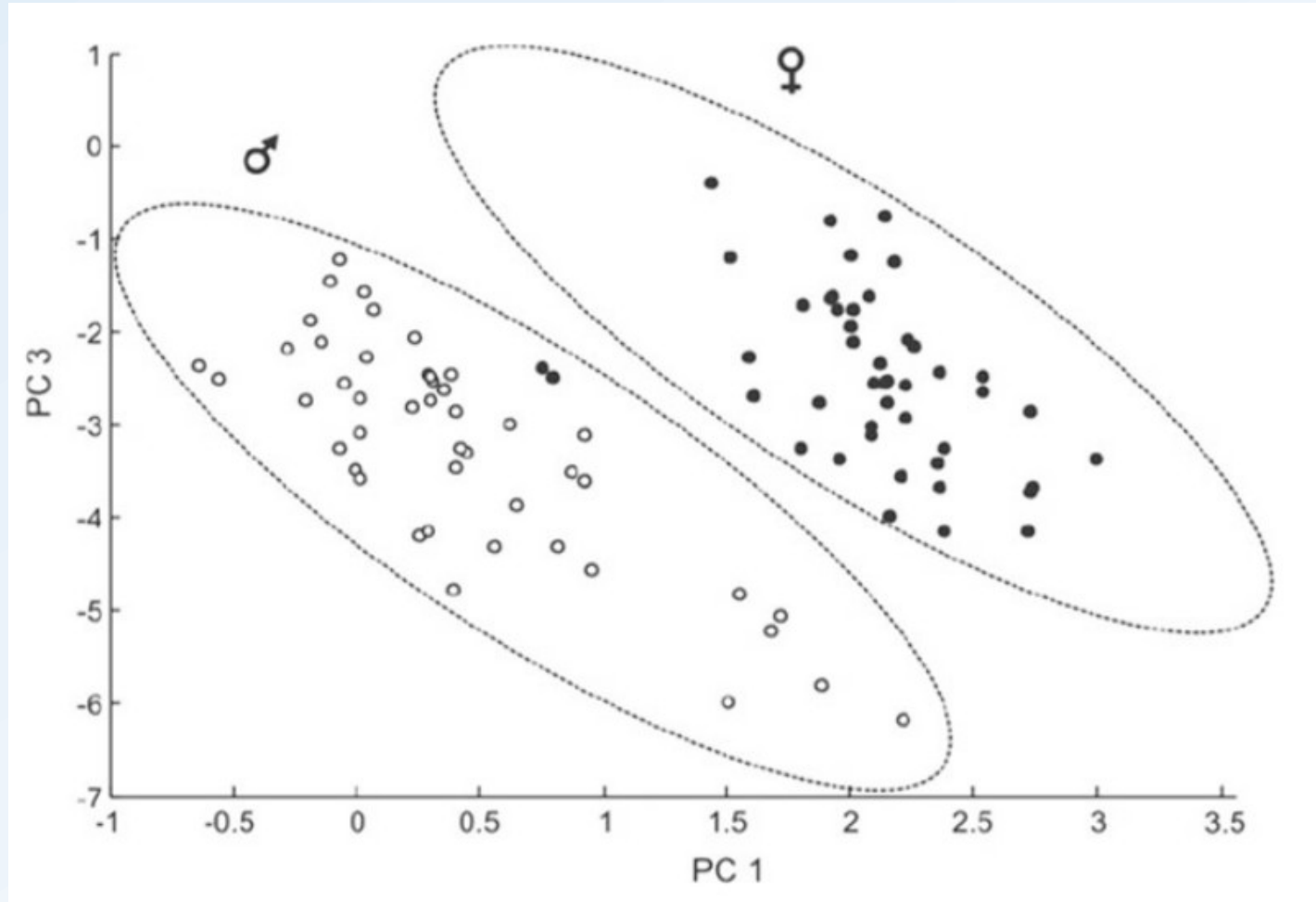
can make spectral analysis no longer a mere provider of analytical data, but a provider of chemical information as well as a direct participant and solver of chemical issues. For example, spectra with pattern recognition methods can accurately identify authentic products such as drugs, food, and cosmetics, as well, can diagnose early stage of cancer, identify sources of oil spills.



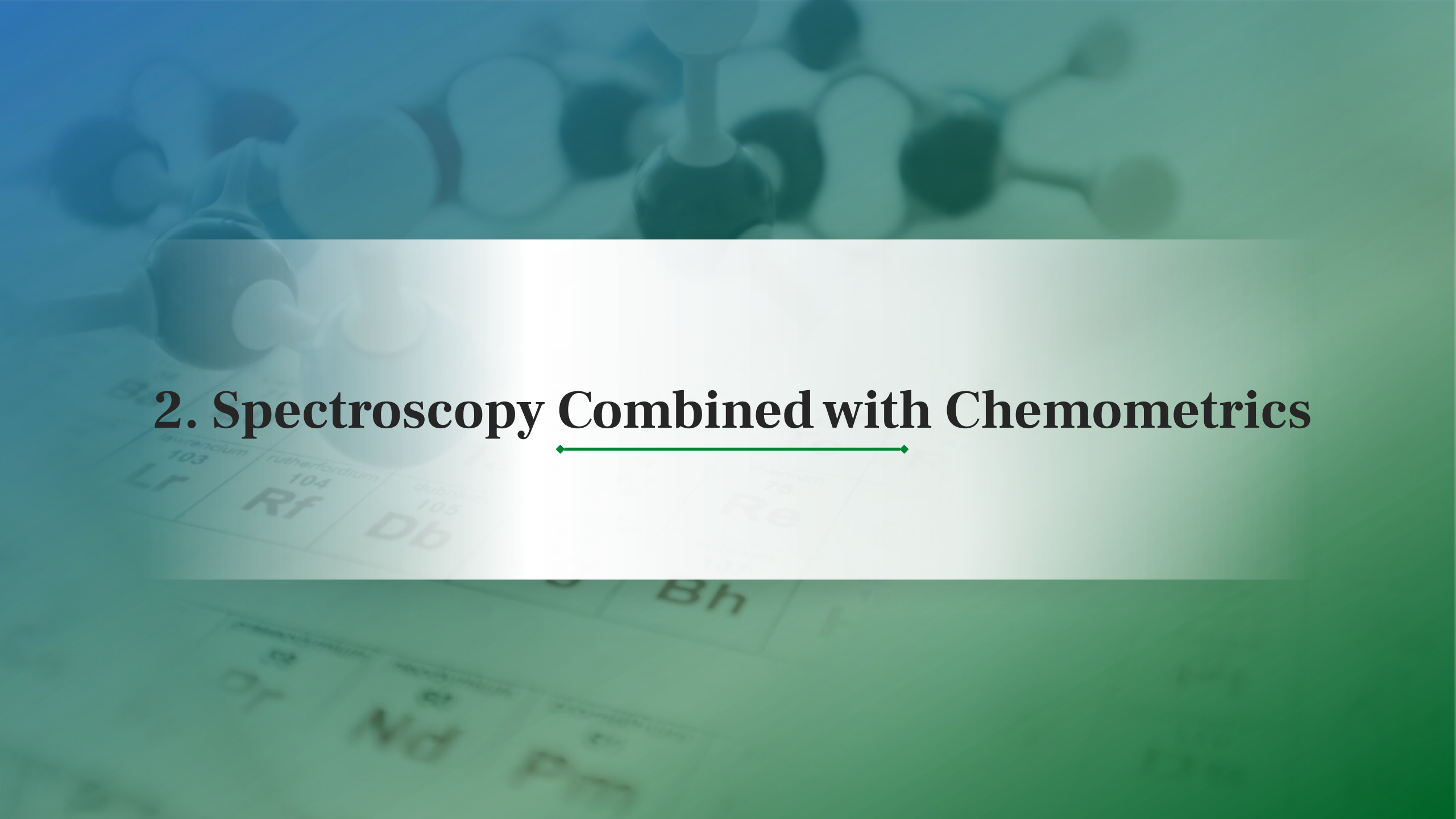
- Mid-infrared spectra of the root-tip material of different gender bird feathers

Spectra of males and females cannot be identified by the traditional characteristic peak method, because they all reflect the functional groups in proteins, nucleic acids, phospholipids, carbohydrates, and ribose.

Necessity of Chemometrics

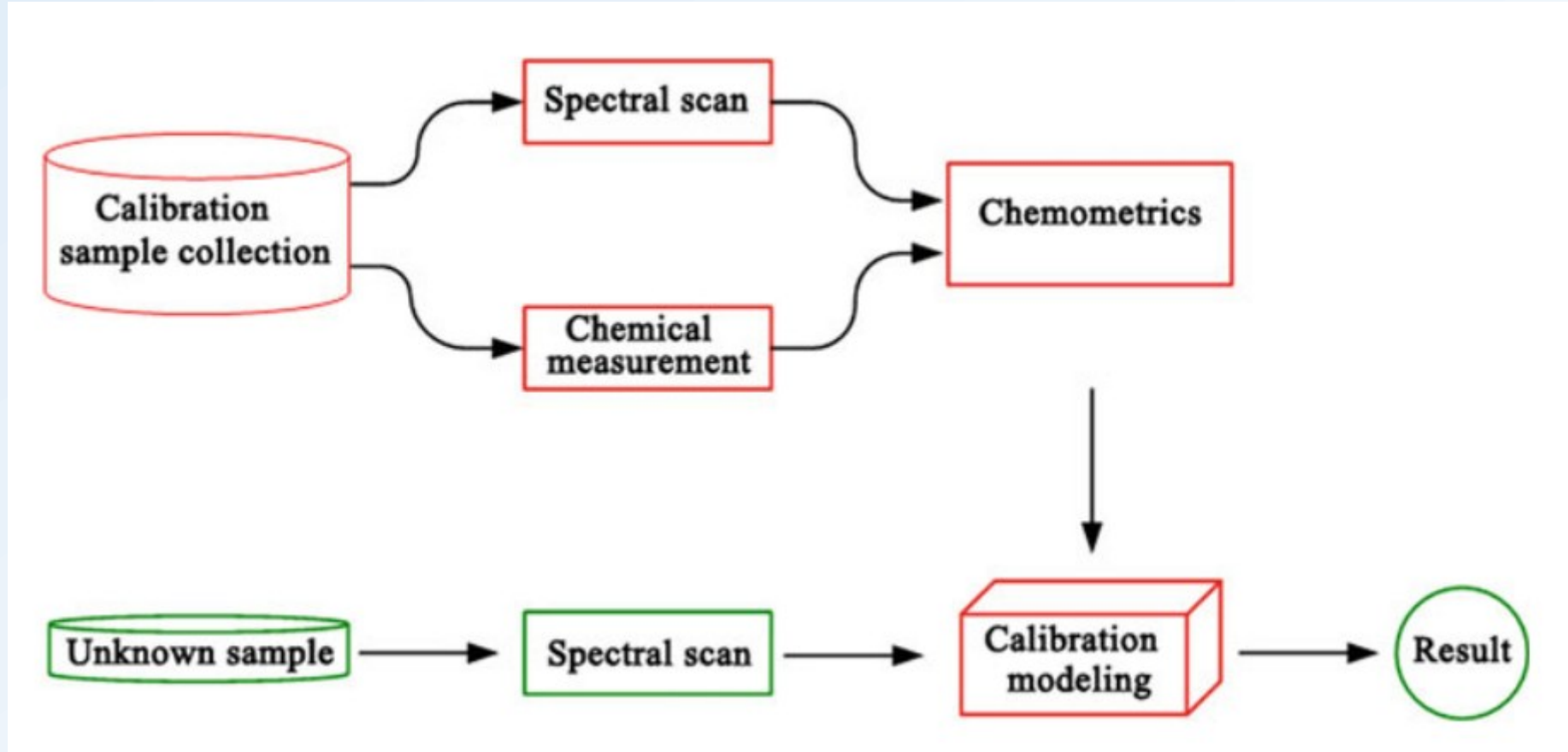


- After extracting the scores of the first and third principal components, the gender of the bird can be clearly distinguished by PCA!

The background features a blurred molecular model with white and grey spheres connected by rods, overlaid on a portion of the periodic table. The periodic table shows elements like Lr (Lawrencium, 103), Rf (Rutherfordium, 104), Db (Dubnium, 105), Re (Rhenium, 75), Bh (Bohrium, 107), Pt (Platinum, 78), and Nd (Neodymium, 60).

2. Spectroscopy Combined with Chemometrics

Paradigm for Spectroscopy Combined with Chemometrics



- Process of establishment of calibration model and prediction of unknown sample

Establishment of Calibration Model



1. Collection of calibration samples ()

There are two requirements for calibration samples. One is that the sample should be **representative**. Its composition should include all the chemical components contained in the sample to be predicted in the future, and its variation range should be greater than that of the corresponding property of the sample to be predicted. Specifically, the variation range is usually **greater than five times** the reproducibility of the reference method, and it is evenly distributed throughout the range. For example, if the reproducibility of the gasoline octane number determined by the standard method is 0.7 units, then variation range of the calibration sample is at least 3.5 units. The second requirement is that the number should be **adequate enough** to effectively extract the mathematical relationship between the spectra and the components to be predicted. For a simple test system, **at least 60** representative samples are required. For a complex system, at least **over one hundred** of representative samples are required.

For collection of natural samples, a variety of influencing factors should be considered. For example, when collecting crop samples, it should include samples of different climates, growing conditions, varieties, textures, and harvest seasons, etc. Online chemical testing should include samples under various process conditions, such as raw materials, temperature, pressure, and catalysts, etc..

Storage, Record, Classification.

2. Acquisition of spectra ()

For near-infrared spectroscopy (NIRS), modes of transmission, diffuse reflection, and diffuse transmission can be selected according to the different objects. Even the same diffuse reflection method, there are different measurement accessories like integrating spheres, diffuse reflection probes, etc. **Thus, the optimal selection of acquisition conditions and standardized measurement are the core content of spectra collection.** The spectra acquisition to be optimized mainly include temperature, optical path, resolution, number of spectral accumulations, and wavelength range, as well as, sample pretreatments such as milling of solid samples, extraction of liquid samples, or fruit slices, etc. In most cases, the samples used for NIRS measurement do not require any pretreatment.

To obtain uniformly measured spectra, **standardized collection** of spectra is very important, that is, spectral measurement conditions of all samples in the same calibration set should be as consistent as possible. Plus, sampling (such as sample inhomogeneity issues) and loading (such as the density of solid particles, the direction of liquid cuvettes, the orientation of single grains or fruits, etc.) should also be standardized.

3. *Selection of calibration sample ()*

Samples that are analyzed in the laboratory usually have thousands of inspections in a few months, but it is possible that more than 80% of these samples are duplicate samples. So, **it is necessary to select the representative samples to establish a calibration model.** It can not only increase the speed of modeling but also reduce the storage space of the library. Furthermore, when encountering samples outside the model boundaries, fewer samples can increase the range of application of the model and facilitate model update and maintenance. **Plus, the cost will be huge.** PCA is usually performed on the spectra of all calibration samples, and then a certain number of representative samples are selected according to their distribution in the principal component space (PCs), such as the commonly used K-S method. When selecting calibration samples, attention should be paid to the outlier samples. In the spatial distribution of PCs, these outliers are significantly different from others, which may contain other components or the extreme concentrations.

4. *Measurement of reference method ()*

The accuracy of the reference data has a greater impact on the prediction of the quantitative model. Therefore, most of the reference data used in modeling are measured by standard methods or conventional analytical methods. If necessary, the accuracy and repeatability of these conventional methods should be evaluated. To obtain the high-accuracy reference data, sometimes it is necessary to take the average value from multiple measurements, use the same instrument and skilled operators to ensure the calibration as much as possible. Finally, the sample used for reference measurement must be the same as that used for spectral collection, and reference data and spectrum of sample should be tested as soon as possible after sampling, so as not to affect the accuracy of the calibration model due to changes in sample composition.

5. *Establishment of calibration* ()

The sequence of calibration is roughly as follows:

- ① Formation a calibration matrix by using spectra and corresponding reference values;
- ② Mathematical transformation of the spectral data (i.e., preprocessing), such as derivative, wavelet transform, multiplicative scatter correction, mean-centering, etc.;
- ③ Selection of spectral variables (intervals), such as correlation coefficients, genetic algorithms, etc.;
- ④ Obtaining a quantitative calibration model by performing regression to preprocessing spectral and property values by PCR, PLS, or ANN. In the process, parameters such as the number of derivative points and the number of optimal PCs need to be determined;
- ⑤ Removal of outliers. An outlier refers to a sample whose predicted value obtained by the interactive validation is significantly different from its actual value;
- ⑥ Reestablish the model. After removing the outliers from the calibration set, the same calibration parameters are used to perform the regression again, and then repeat until a satisfactory quantitative model is obtained.

Establishment of Calibration Model



6. Validation of model ()

After the model being built, a set of known samples (validation set) need to be used to validate the accuracy, stability, robustness, and transferability of the model. Validation set samples should contain all the components contained in the sample to be predicted, concentration range of which should cover at least 95% of that in the calibration set, with the uniform distribution. Plus, samples in the validation set should be enough for statistical testing, usually no less than 28 samples are required. The robustness of a model refers to its performance against external interference factors. These influencing factors mainly include the replacement of the same type of test devices (such as cuvettes, optical fiber probes, integrating spheres, etc.), changes in the degree of fiber bending, replacement of light sources, replacement of reference materials (such as ceramic chips or barium sulfate powder, etc.), changes in sample loading conditions, changes in temperature (ambient and sample temperature), and changes in the physical state of particles (such as grain moisture content, changes in polymer particle size, residual solvents), etc..

The transferability of model mainly depends on the hardware differences between the instrument systems, and its essence is the replaceability of the spectrometer and its key components (optical systems such as interferometers). Transferability of model directly affects the generalization performance of the analytical method for the user. If the spectrometer of the same manufacturer does not have the transferability of the model, it is very difficult to share abundant model resources. Usually, different instruments have significant system deviations, and the spectra need to be corrected, that is, the **calibration transfer**, in order to get consistent results. However, there are also manufacturers that can achieve consistency between instruments, and their calibration models can be directly used for the same type of spectrometer without any modification. That is, the model data is directly copied and transmitted (**Calibration transport**).

7. *Applicability criterion*()

Since it is impossible to establish a calibration model that covers all unknown samples, it is particularly important and necessary to establish the applicability criterion for the model. Before performing prediction on unknown samples, the validity and accuracy of results can be guaranteed only if the sample to be tested is within the range covered by the model.

Generally, there are three criteria can be used to ensure the applicability of the model. One is the **Mahalanobis distance** (□ □ □ □). If the Mahalanobis distance of tested sample is farer than the maximum distance of the calibration set, it indicates that the concentration of certain components in the tested exceeds the range of that of the calibration set. The second is the **spectral residual** (□ □ □ □). If the spectral residual of the tested is over the specified threshold, it means the tested contains components that are not in the calibration set sample. The third is the nearest neighbor distance. If the minimum value of the distance between the tested and all calibration samples (**the nearest neighbor distance**) is over the specified threshold, it is implied that the tested falls into a place where the distribution of the calibration set is relatively sparse, and the accuracy of the prediction result will be suspected. **Establishment of a robust, reliable, and highly accurate calibration model is the key to the success of the analytical method.**

Establishment of Calibration Model



The influencing factors:

- (1) ***Influence of the calibration sample.*** It includes the representativeness, quantity, range, and distribution; storage; uniformity (such as the particle size, sprout rate, water content, color, and impurities of agro-products); preprocessing (such as crushing, slicing and extraction, etc.); accuracy of the reference data, etc.
- (2) ***Influence of acquisition condition.*** It includes spectral range, resolution, acquisition method (such as diffuse reflectance accessory is integrating sphere or fiber optic probe, choice of background material, choice of optical path in transmission method, etc.), temperature, uniformity, and consistency of sampling and loading, etc. Each type of sample (clear liquid, turbid suspension, milled powder, or coarse particles) has its most suitable measuring accessories.
- (3) ***Influence of chemometrics.*** It includes spectral pretreatments and their parameters, selection of wavelength variables, calibration methods, and parameters (linear/nonlinear methods, under/over-fitting judgments, and removal of outliers, etc.).
- (4) ***Influence of instrument (repeatability and long-term stability).*** It includes effective wavelength range, resolution, S/N ratio, baseline stability, wavelength accuracy and repeatability, absorbance accuracy and repeatability, temperature application range, and resistance to voltage fluctuations, etc.

Routine Analysis



After validation, the tested samples can be routinely analyzed. Spectra of the predicted samples should be collected in accordance with the **measurement of calibration sample**, such as resolution, background, sample and ambient temperature, loading method, and pretreatment method (milling), etc. Applicability of model should be judged before performing routine analysis on the tested sample. If the applicability criterion exceeds the threshold, the model is not suitable for the quantitative analysis of sample.

Models and instruments need to be tested regularly during routine analysis. It is also called the assurance or quality control of analysis. Routine analysis can be conducted in the following ways:

- ① **Actual samples are used for regular verification**, such as 2–3 times a week, and compared with the reference method. The absolute deviation should not exceed the range of reproducibility.
- ② If tested sample can be sealed, about 3–5 representative samples are selected and stored in a sealed enclosure. Routine analysis is done as once every 2 days, **and evaluate by the quality control chart.**
- ③ If the composition of tested sample is simple, **the accuracy can be verified regularly by preparing standard samples.**

If there were inconsistent results in the test, the spectra should be re-collected multiple times, and predictive analysis should be performed to ensure that the spectra are collected correctly, and then the accuracy of the reference data should be checked. If there are still significant differences, the hardware of the spectrometer needs to be fully tested and checked until the cause of the error is located.

Features of the Method

Compared with traditional methods, analytical method combining spectroscopy with chemometrics has the following **significant advantages**:

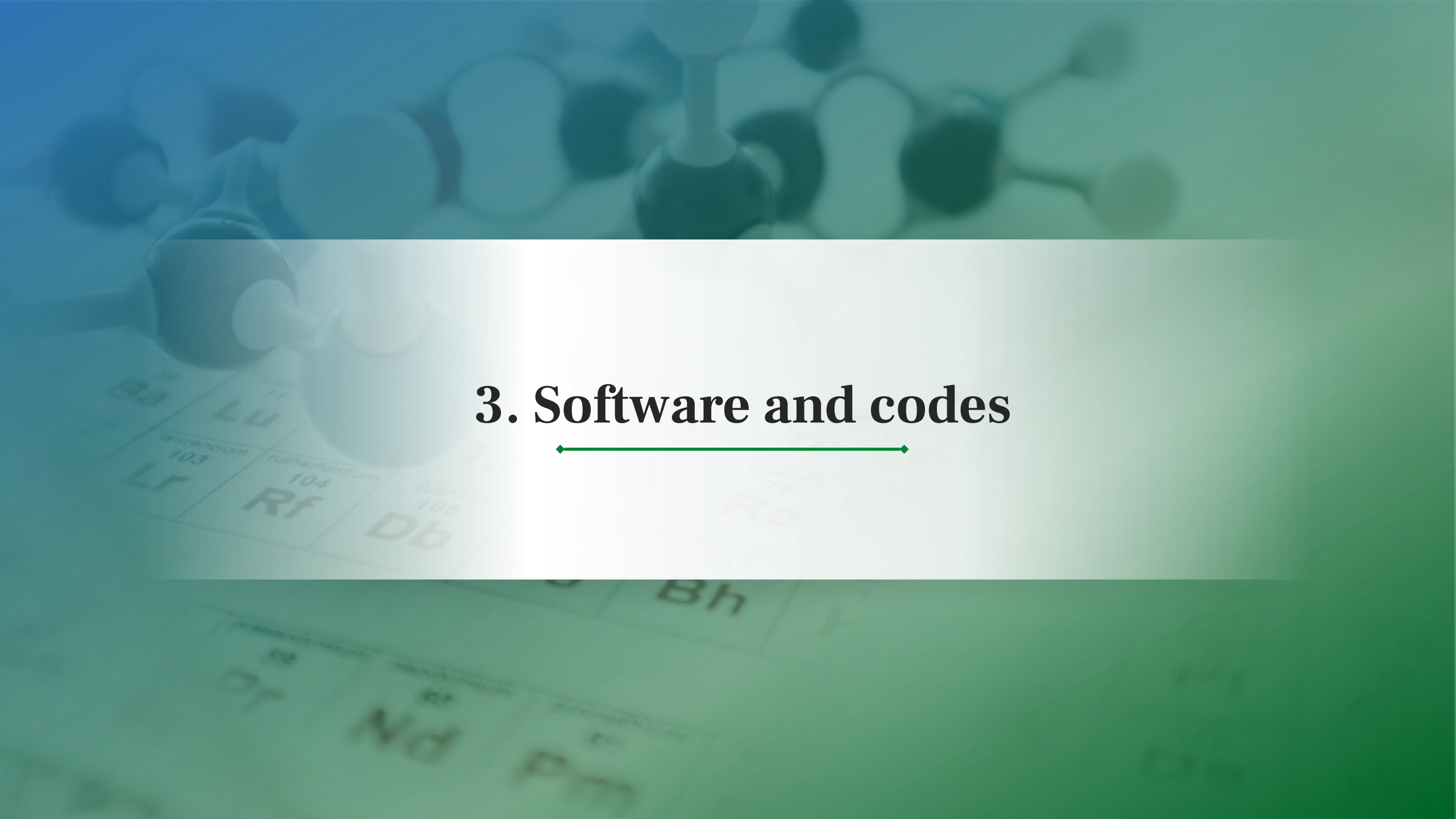
- (1) It can perform non-destructive analysis of complex mixtures of various forms, usually without sample processing. It directly acquires the spectra without chemical reagents, being as an environmentally friendly analysis technology.
- (2) Analytical speed is pretty fast and the efficiency is satisfactory. A spectrum can be used to determine the multiple composition and property data of the sample within a few seconds.
- (3) The repeatability and reproducibility of analysis are generally better than conventional analytical methods.
- (4) Easy to realize on-site rapid analysis.
- (5) The instrument has few wearing parts and consumables, and low maintenance.
- (6) Most online analyzers can use optical fiber transmission technology, which is suitable for harsh environments.

The limitations are mainly as follows:

- (1) Quantitative and qualitative analyses almost completely depend on the calibration model that often needs to be established separately for different sample types with a lot of resources. Therefore, this method is not suitable for the small amount samples, and not suitable for analysis that can be completed easily by conventional methods.
- (2) Establishment of the calibration model is not done once and for all. In practice, calibration model needs to be continuously expanded and maintained according to the composition change of the tested sample.
- (3) Calibration model requires long-term stability of the spectrometer without the significant change of optical components in the instrument.

Take home messages

- 1. When using chemometrics, a deep understanding and mastery of the field involved in the problem or relevant chemical background should be possessed firsts.*
- 2. When dealing with practical problems, it is necessary to choose the appropriate chemometric method according to the specific case, instead of using the latest or the most complicated method. In fact, some basic chemometric concepts can address many application problems.*



3. Software and codes



Contents

Any software for chemometrics?

How to perform traditional chemometrics by python?

- **Correlation coefficient**
- **Normalization**
- **Smoothing**
- **MLR, PCA, PCR**
- **PLS**

➤ Correlation coefficient □ □ □ □ □ □

Correlation relationship is uncertain relationship, such as the relationship between height and weight, and the relationship between the teachers and graduation rates, which does exist a certain relation between variables, but not the one-to-one relationship. This relationship can be described by correlation (correlation coefficient R or determination coefficient R^2)

$$R = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Pearson correlation coefficient

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

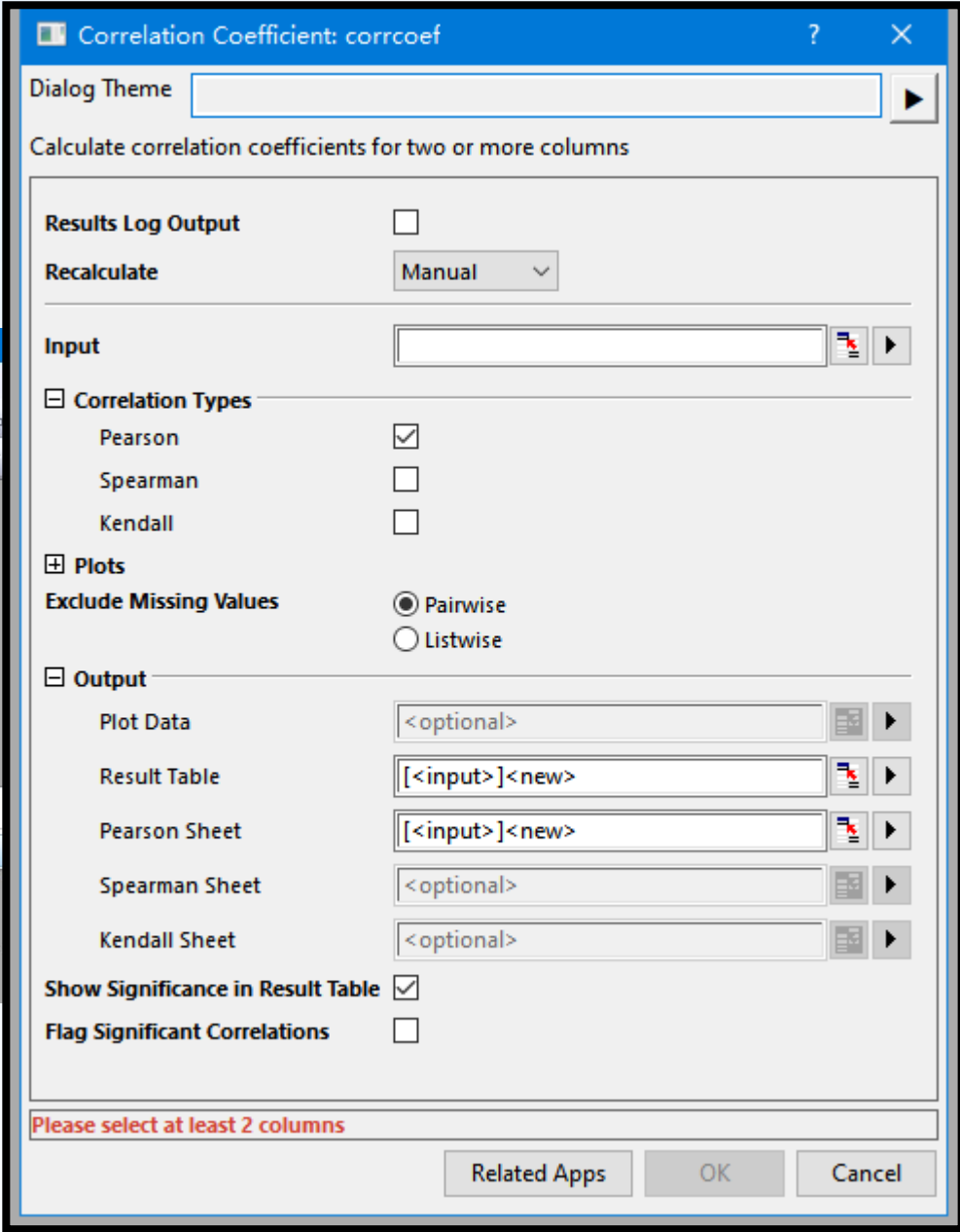
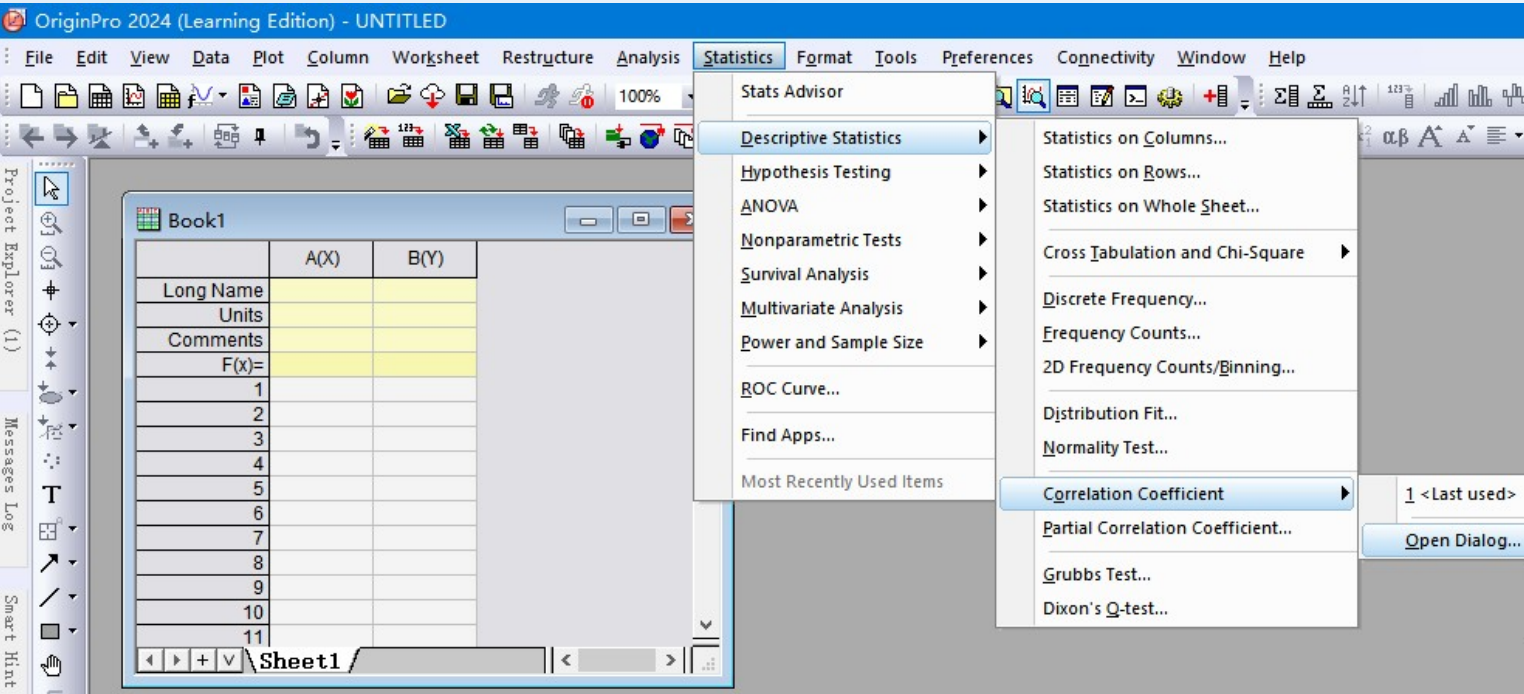
Spearman rank correlation coefficient

$$\tau = \frac{(\text{number of concordant pairs}) - (\text{number of discordant pairs})}{(\text{number of pairs})} = 1 - \frac{2(\text{number of discordant pairs})}{\binom{n}{2}}$$

Kendall rank correlation coefficient

Software and codes

➤ Correlation coefficient



➤ Correlation coefficient

numpy.corrcoef

`numpy.corrcoef(x, y=None, rowvar=True, bias=<no value>, ddof=<no value>, *, dtype=None)`

Return Pearson product-moment correlation coefficients.

[\[source\]](#)

Please refer to the documentation for `cov` for more detail. The relationship between the correlation coefficient matrix, R , and the covariance matrix, C , is

$$R_{ij} = \frac{C_{ij}}{\sqrt{C_{ii}C_{jj}}}$$

The values of R are between -1 and 1, inclusive.

Parameters: `x` : *array_like*

A 1-D or 2-D array containing multiple variables and observations. Each row of x represents a variable, and each column a single observation of all those variables. Also see `rowvar` below.

`y` : *array_like, optional*

An additional set of variables and observations. y has the same shape as x .

`rowvar` : *bool, optional*

If `rowvar` is True (default), then each row represents a variable, with observations in the columns. Otherwise, the relationship is transposed: each column represents a variable, while the rows contain observations.

`bias` : *_NoValue, optional*

Has no effect, do not use.

If we add another set of variables and observations `yarr`, we can compute the row-wise Pearson correlation coefficients between the variables in `xarr` and `yarr`.

```
>>> yarr = rng.random((3, 3))
>>> yarr
array([[0.45038594, 0.37079802, 0.92676499],
       [0.64386512, 0.82276161, 0.4434142 ],
       [0.22723872, 0.55458479, 0.06381726]])
>>> R2 = np.corrcoef(xarr, yarr)
>>> R2
array([[ 1.          ,  0.99256089, -0.68080986,  0.75008178, -0.934284  ,
        -0.99004057],
       [ 0.99256089,  1.          , -0.76492172,  0.82502011, -0.97074098,
        -0.99981569],
       [-0.68080986, -0.76492172,  1.          , -0.99507202,  0.89721355,
         0.77714685],
       [ 0.75008178,  0.82502011, -0.99507202,  1.          , -0.93657855,
        -0.83571711],
       [-0.934284  , -0.97074098,  0.89721355, -0.93657855,  1.          ,
         0.97517215],
       [-0.99004057, -0.99981569,  0.77714685, -0.83571711,  0.97517215,
         1.          ]])
```

```
>>> import numpy as np
>>> # total collagen (mg/g dry weight of liver)
>>> x = np.array([7.1, 7.1, 7.2, 8.3, 9.4, 10.5, 11.4])
>>> # free proline (μ mole/g dry weight of liver)
>>> y = np.array([2.8, 2.9, 2.8, 2.6, 3.5, 4.6, 5.0])
```

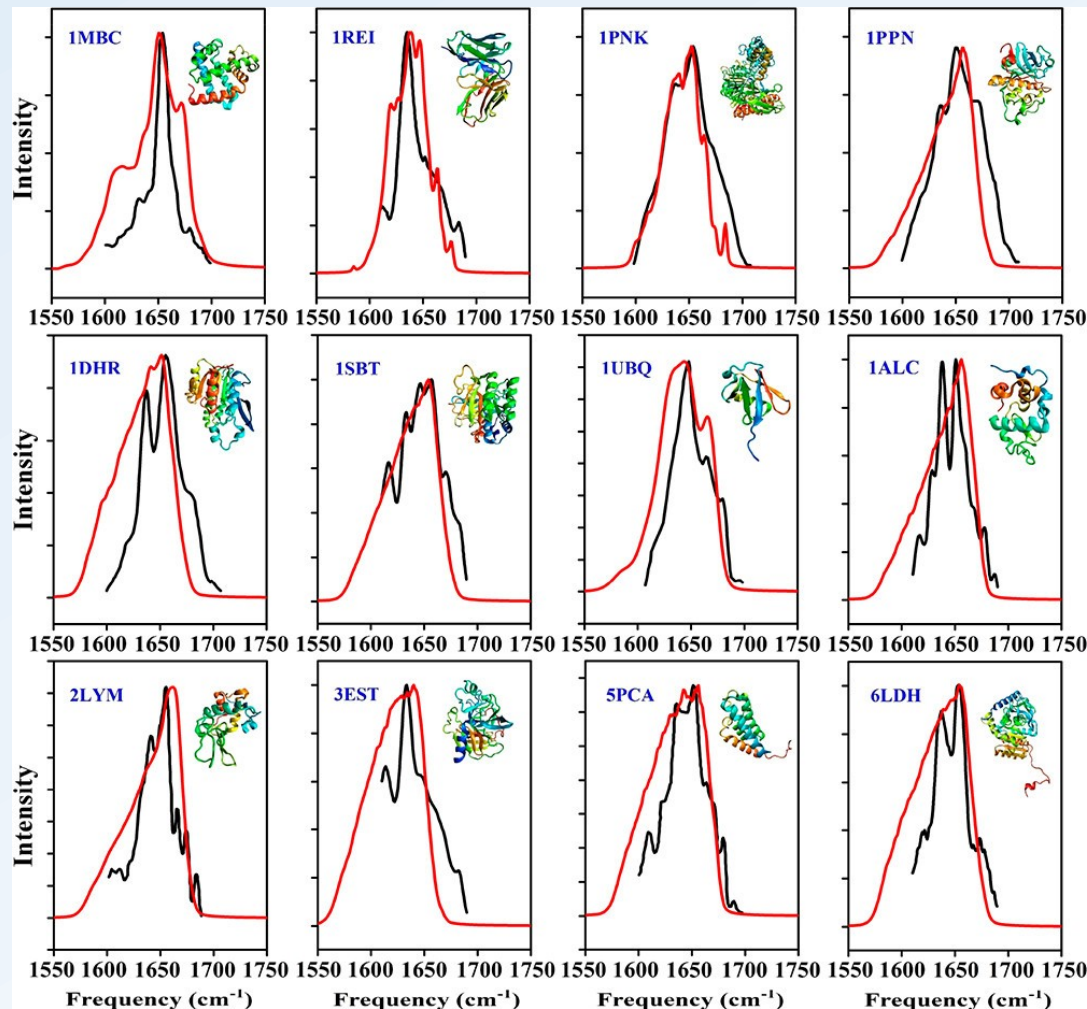
These data were analyzed in [4] using Spearman's correlation coefficient, a statistic sensitive to monotonic correlation between the samples.

```
>>> from scipy import stats
>>> res = stats.spearmanr(x, y)
>>> res.statistic
0.7000000000000001
```

Software and codes



➤ Correlation coefficient



They utilized Spearman rank correlation coefficients, and all correlation coefficient is over 0.9!

Thinking: Is it reliable?

J. Am. Chem. Soc. 2020, 142, 45, 19071–19077

Software and codes

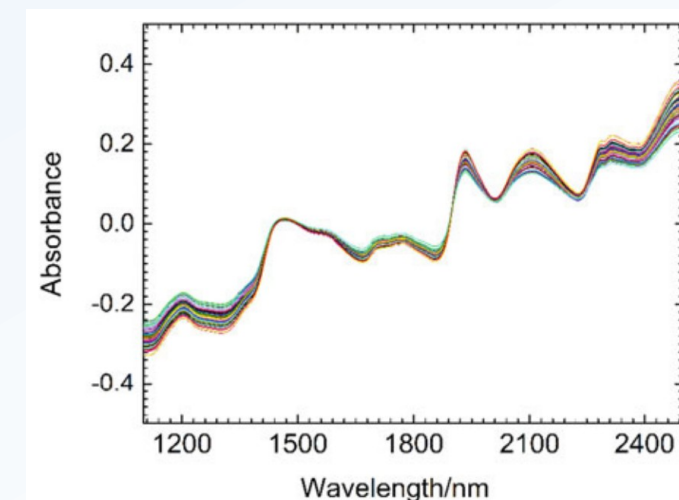
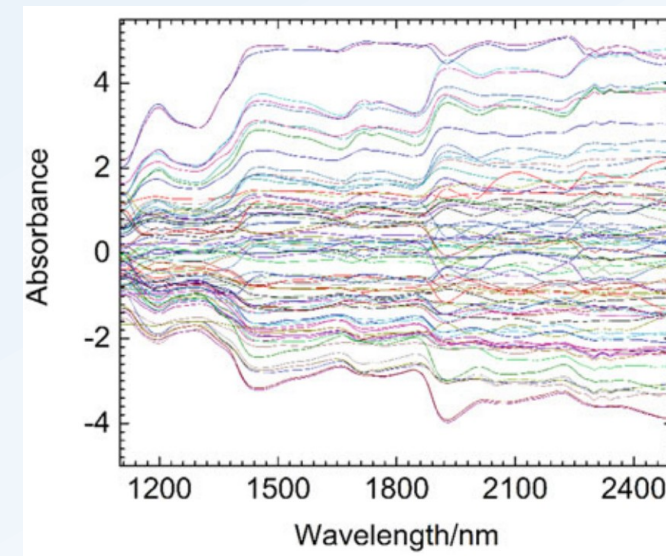


➤ Normalization □ □ □ □ □

Normalization has many algorithms, such as area normalization, maximum normalization, and average normalization. In spectral analysis, the most commonly used normalization is vector normalization. For a spectrum $\mathbf{x}$ ($1 \times m$), the vector normalization algorithm is as follows:

$$\mathbf{x}_{\text{normalized}} = \frac{\mathbf{x} - \bar{x}}{\sqrt{\sum_{k=1}^m x_k^2}}$$

where m is the number of wavelength points, $k = 1, 2, \dots, m$. This is often used to correct spectral changes caused by small light path differences.



NOTE: Understanding the mechanism and your purpose to do this and what can you obtain!

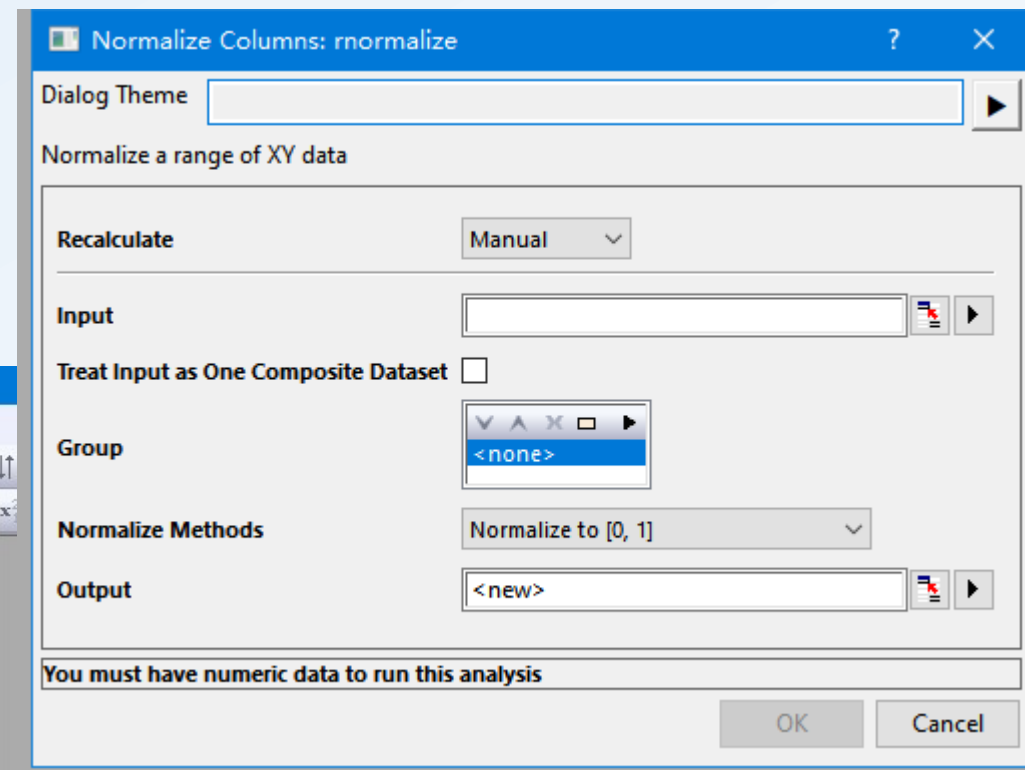
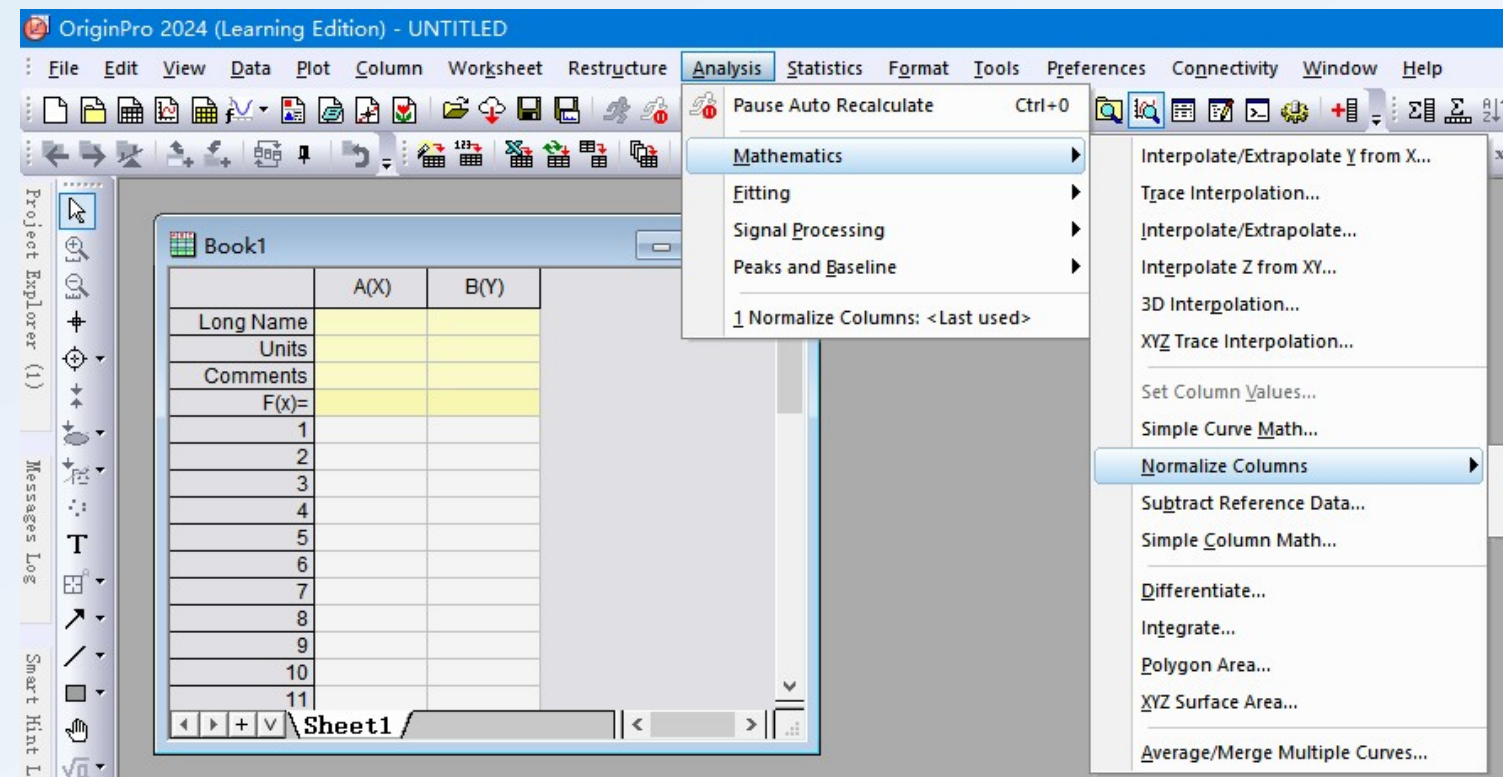
➤ Normalization ☐ ☐ ☐ ☐ ☐

Name	Formula	Use
Standard score	$\frac{X - \mu}{\sigma}$	Normalizing errors when population parameters are known. Works well for populations that are normally distributed ^[2]
Student's t-statistic	$\frac{\hat{\beta} - \beta_0}{\text{s. e.}(\hat{\beta})}$	the departure of the estimated value of a parameter from its hypothesized value, normalized by its standard error.
Studentized residual	$\frac{\hat{\varepsilon}_i}{\hat{\sigma}_i} = \frac{X_i - \hat{\mu}_i}{\hat{\sigma}_i}$	Normalizing residuals when parameters are estimated, particularly across different data points in regression analysis .
Standardized moment	$\frac{\mu_k}{\sigma^k}$	Normalizing moments, using the standard deviation σ as a measure of scale.
Coefficient of variation	$\frac{\sigma}{\mu}$	Normalizing dispersion, using the mean μ as a measure of scale, particularly for positive distribution such as the exponential distribution and Poisson distribution .
Min-max feature scaling	$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$	Feature scaling is used to bring all values into the range [0,1]. This is also called unity-based normalization. This can be generalized to restrict the range of values in the dataset between any arbitrary points a and b , using for example $X' = a + \frac{(X - X_{\min})(b - a)}{X_{\max} - X_{\min}}$.

Software and codes



➤ Normalization



➤ Normalization ☐ ☐ ☐ ☐ ☐

$$z = \frac{x - \mu}{\sigma}$$

where:

μ is the **mean** of the population,

σ is the **standard deviation** of the population.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Hint:

`numpy.mean( )`

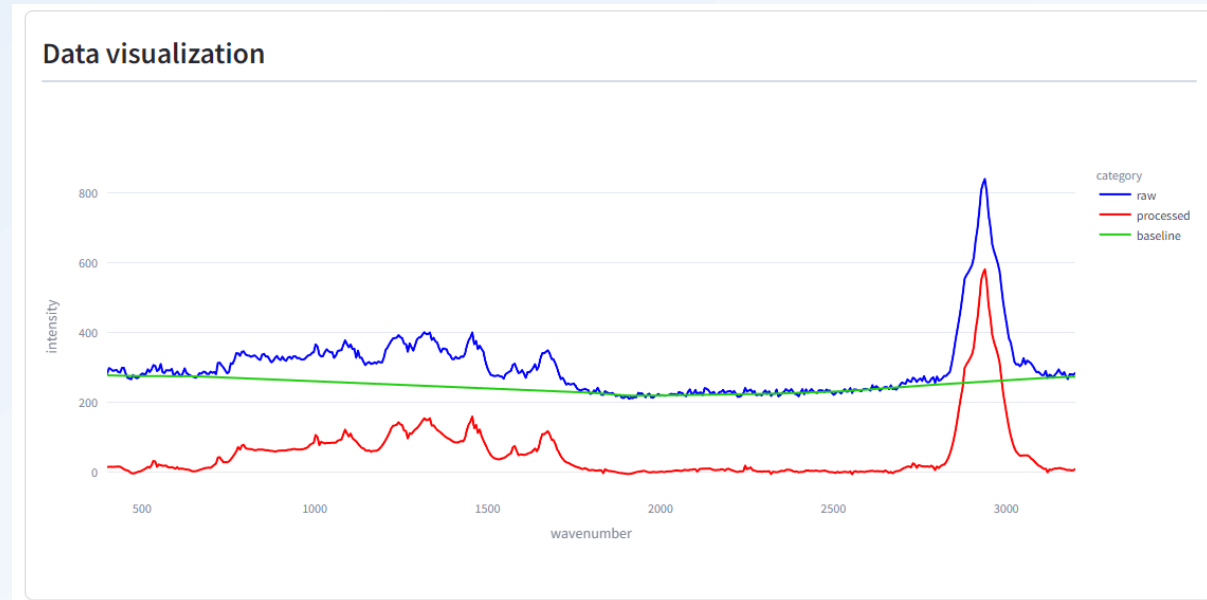
`numpy.std( )`

`numpy.max( )`

`numpy.min( )`

➤ Smoothing □ □ □ □

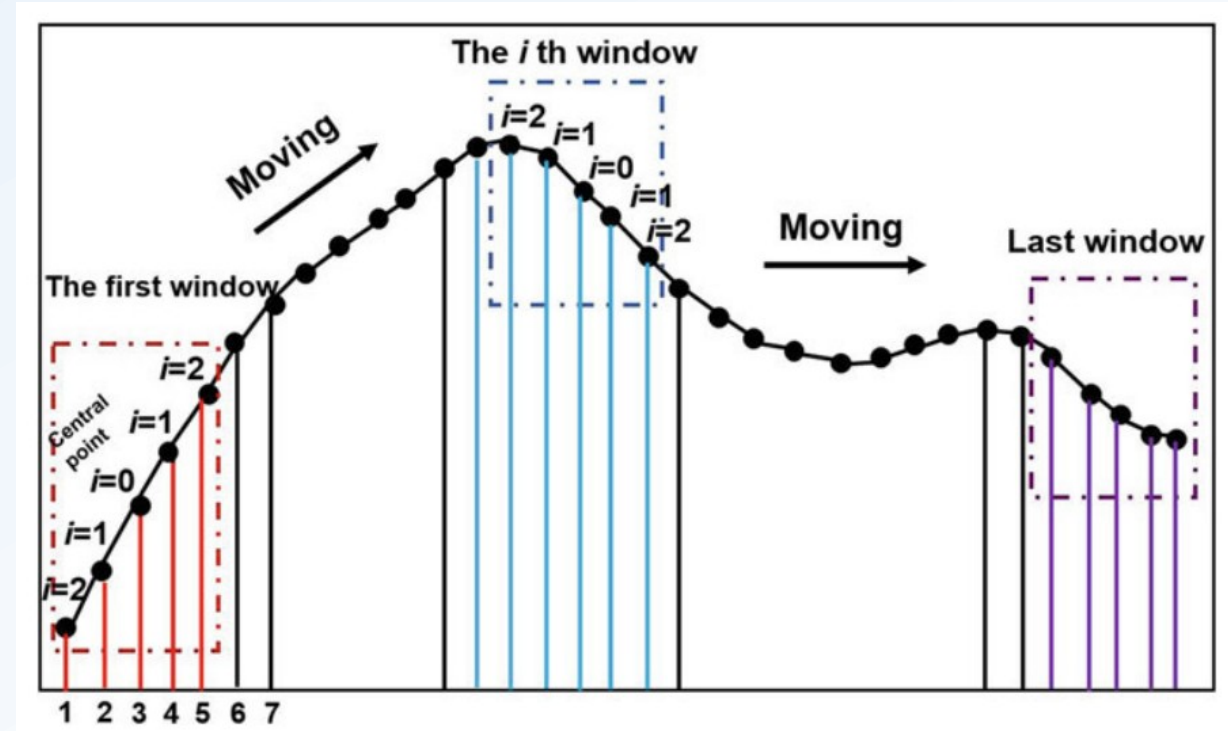
The spectral signals obtained by the spectrometer contain both useful information and random error, i.e., noise. Signal smoothing is one of the most commonly used de-noising methods. It mainly takes the average of multiple measurements to reduce the noise and improve the signal-to-noise ratio when the noise contained in the spectrum is zero mean random white noise. The commonly used signal smoothing methods are **moving average smoothing** and **Savitzky-Golay convolution smoothing**. Moreover, **Fourier transformation** and **wavelet transformation** can also be used for spectral de-noising.



➤ Smoothing □□□□

The **moving average smoothing** selects a window of smoothing with a certain width ($2w+1$), with an odd number of wavelength points in each window. Then the measured value of k wavelength point is replaced by the central wavelength point k in the window and the mean measured $\bar{x}_k$ at w before and after k . At last, k from left to right is moved to complete the smoothing of all points.

$$x_{k,\text{smooth}} = \bar{x}_k = \frac{1}{2w+1} \sum_{i=-w}^{+w} x_{k+i}$$

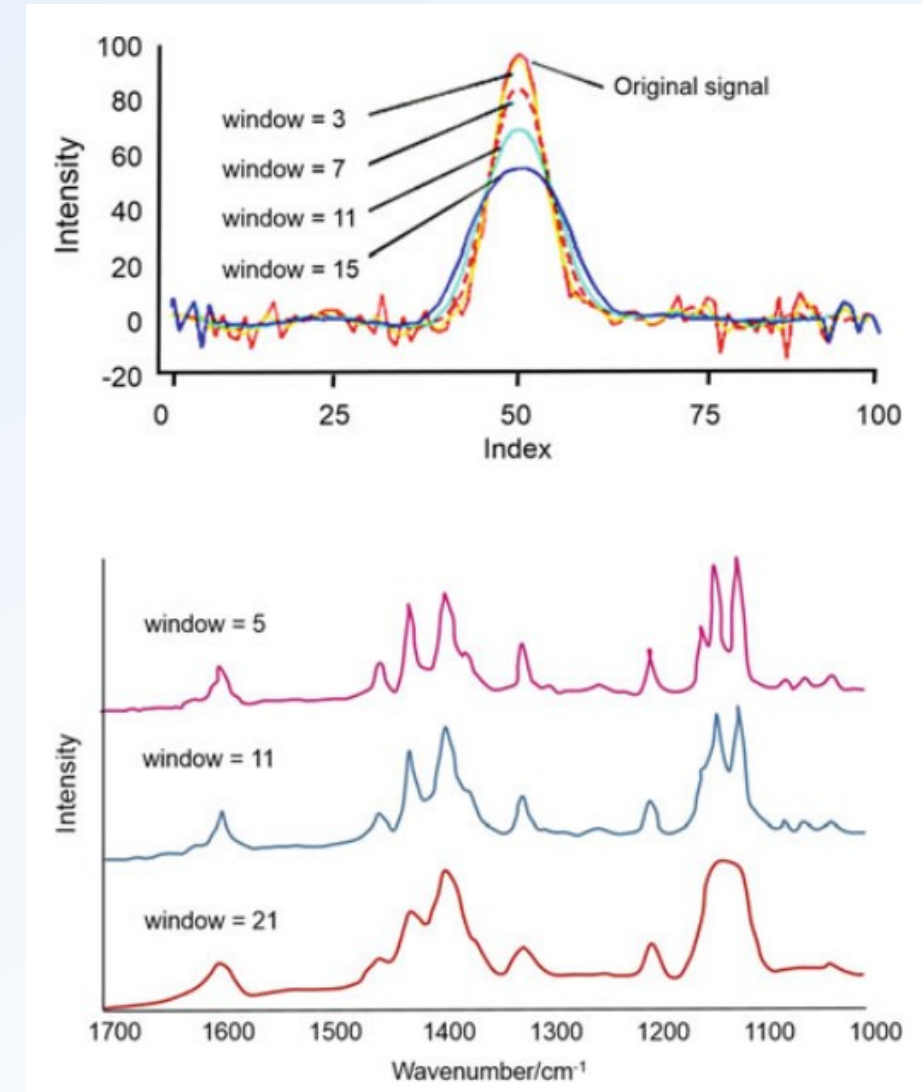


Software and codes



➤ Smoothing □ □ □ □

When using the moving average smoothing method, the **window of smoothing width** is an important parameter. If the window width is too small, the de-noising effect is not perfect. If the window width is too large and its mean is calculated, the useful information is smoothed while smoothing the noise, resulting in the distortion of the spectral signal.



Software and codes



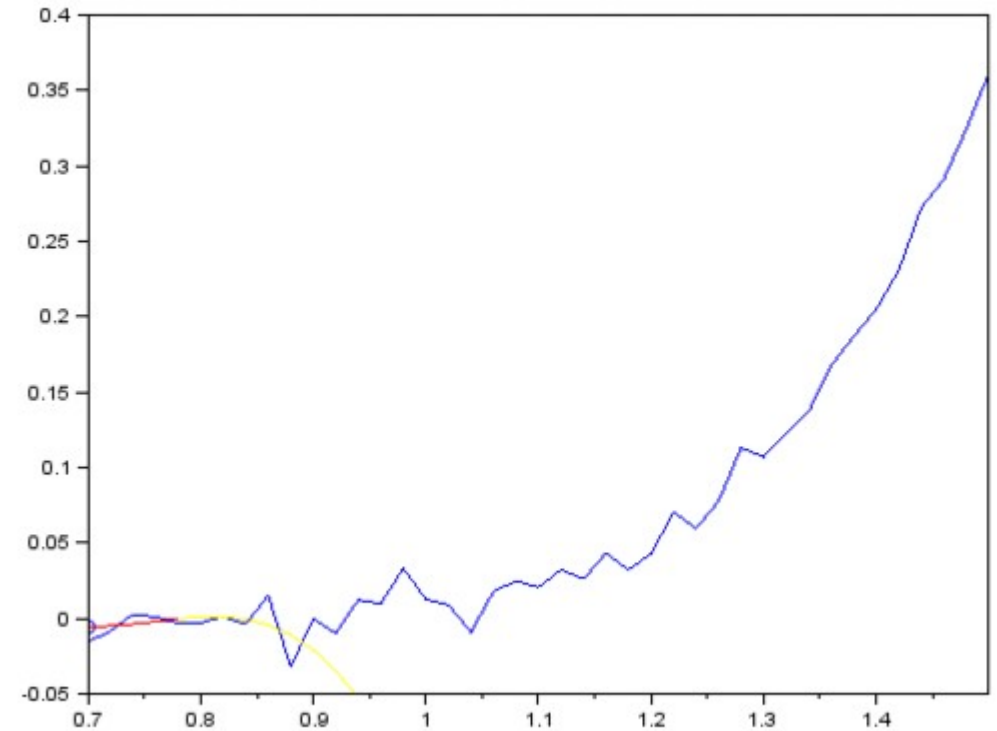
➤ Smoothing □ □ □ □

Savitzky-Golay (S-G) convolution smoothing is also known as polynomial smoothing. The average after smoothing at wavelength k is as follows:

$$x_{k,smooth} = \bar{x}_k = \frac{1}{H} \sum_{i=-w}^{+w} x_{k+i} h_i$$

$$H = \sum_{i=-w}^{+w} h_i$$

Where h_i and H are the smoothing factor and the normalization factor, respectively. The purpose of multiplying each measurement by the smoothing factor h_i is to reduce the effect of smoothing on useful information. x_k can be obtained by using polynomial fit based on the principle of least squares.



Software and codes

➤ Smoothing □ □ □ □

The basic idea of S-G convolution smoothing is similar to moving average smoothing. The difference between them is that S-G convolution smoothing is polynomials least square fitting of the data in the moving window, which is essentially a weighted average with more emphasis on the central role of the central point. S-G convolution smoothing is a widely used de-noising method. The effect of **moving window width** (often referred to number of smoothing points) is significantly lower than the moving average smoothing.

```
from scipy.signal import savgol_filter
```

```
#
```

```
#          SG
```

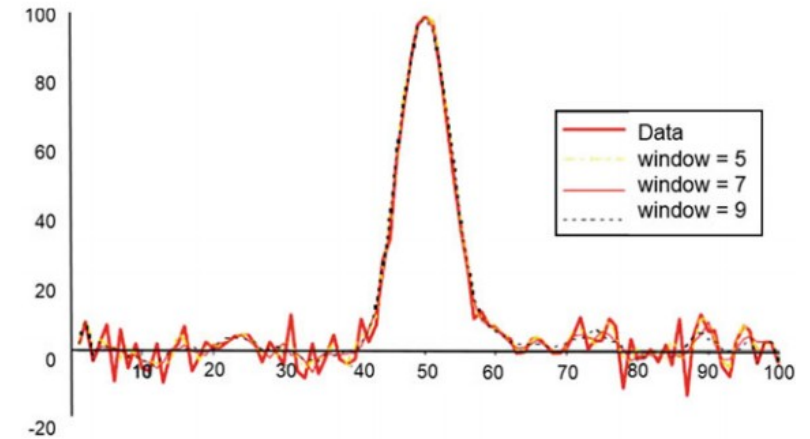
```
window_length = 7
```

```
# SG
```

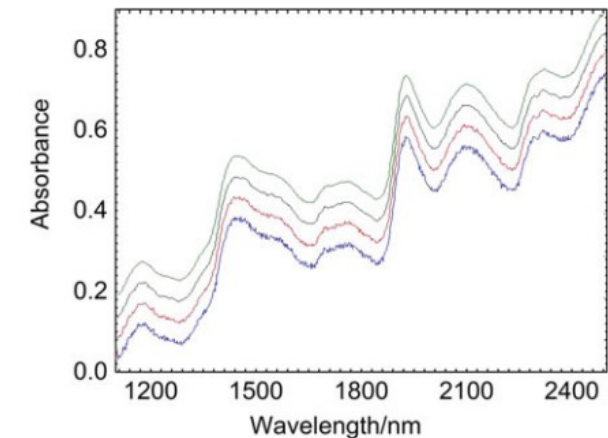
```
poly_order = 3
```

```
# SG
```

```
smoothed_spectrum = savgol_filter(observed_spectrum, window_length, poly_order)
```



Smoothing effects of quadratic polynomial S-G method with different window sizes



Original and smoothing spectra by S-G smoothing method. From bottom to top are the noised spectrum, the smoothing spectra by S-G cubic polynomial with 5 point, 11 point, and 23 point, respectively. An artificial shift was added to absorbance for distinguishing different spectra

Software and codes



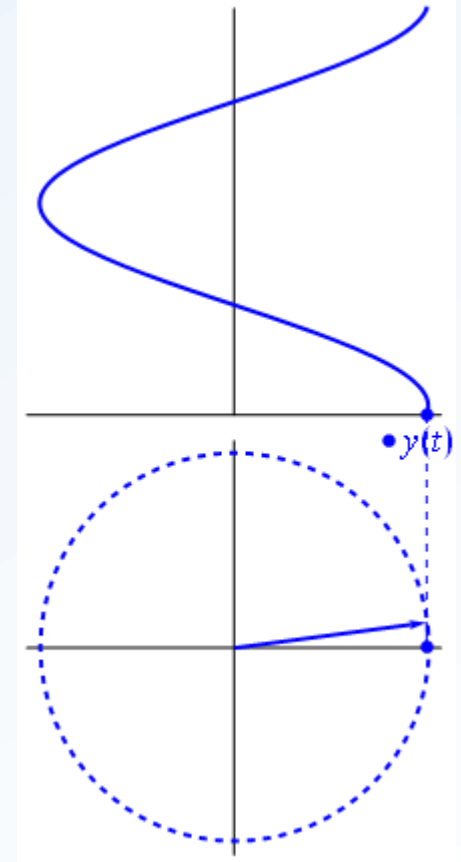
➤ Smoothing □ □ □ □

The Fourier transform () for the spectrum is to decompose the spectrum into the sum of sine waves with different frequencies. Compared with the useful signal, instrument noise has the characteristics of small amplitudes and high frequencies. Therefore, high frequencies are deleted. The original spectrum is reconstructed by inverse Fourier transform from low frequencies. Most spectral noise is eliminated and the spectrum is smoothed by this method.

The basic idea of wavelet transform () for de-noising is similar to the Fourier transform. Wavelet coefficients corresponding to the high-frequency scales obtained by the wavelet transform are removed, and then the spectrum after de-noising is reconstructed.

$$\hat{f}(\xi) \cdot e^{i2\pi\xi x} = Ae^{i\theta} \cdot e^{i2\pi\xi x} = \underbrace{Ae^{i(2\pi\xi x + \theta)}}_{\text{polar coordinate form}} = \underbrace{A \cos(2\pi\xi x + \theta) + iA \sin(2\pi\xi x + \theta)}_{\text{rectangular coordinate form}}.$$

$$[W_{\psi} f](a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} \overline{\psi\left(\frac{x-b}{a}\right)} f(x) dx$$



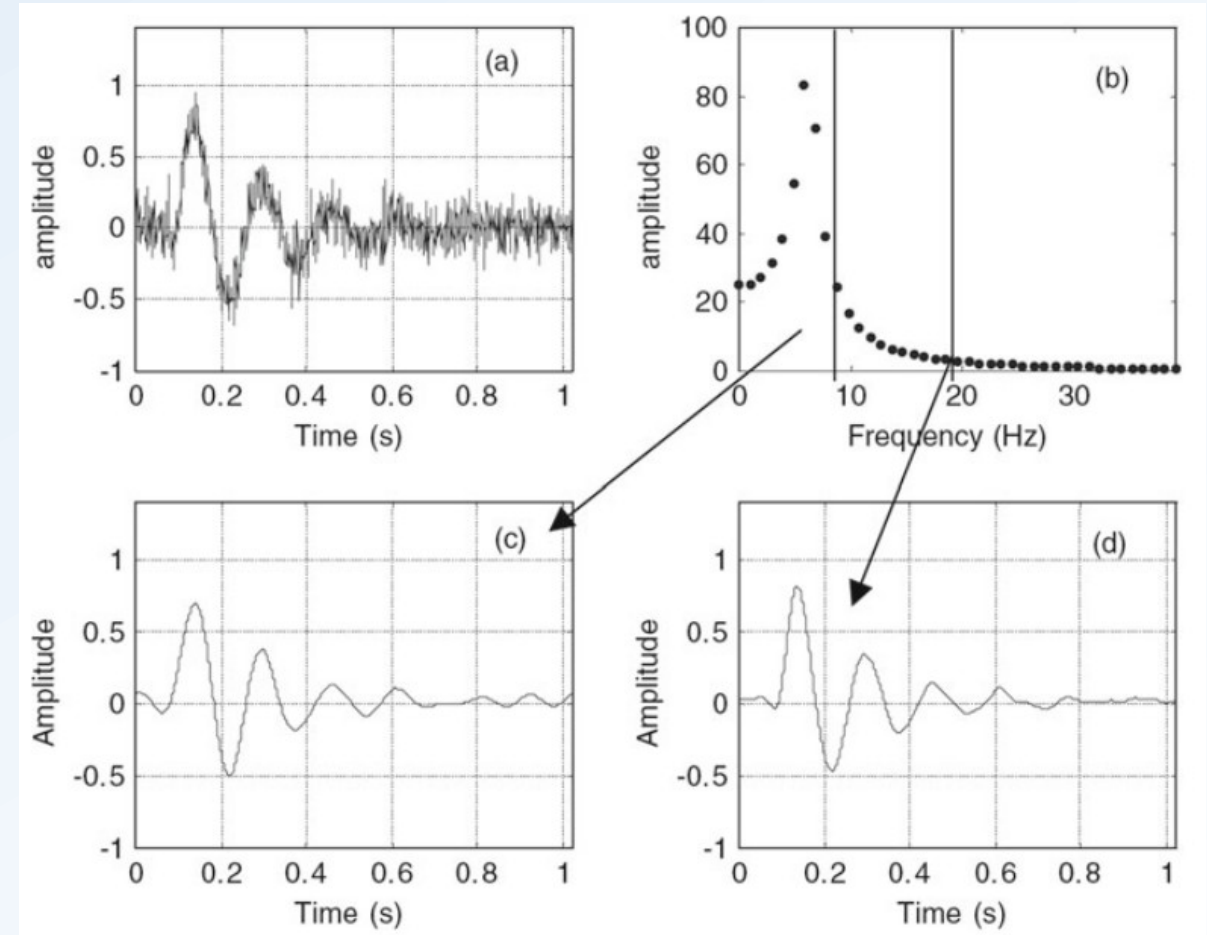
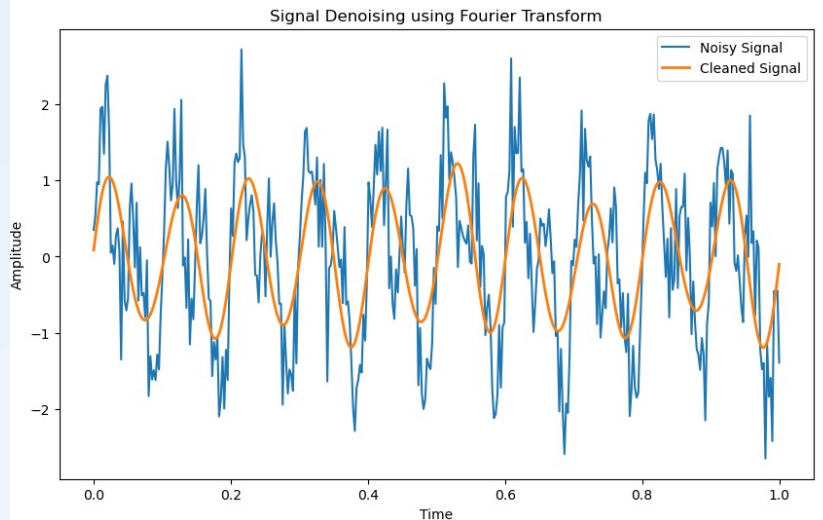
Software and codes

➤ Smoothing □ □ □ □

```
In [20]: import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, ifft

In [27]: # 生成含有噪声的信号
t = np.linspace(0, 1, 400)
x = np.sin(2*np.pi*10*t) + np.sin(2*np.pi*20*t) + np.random.randn(400)*0.1

In [28]: # 进行傅里叶变换
X = fft(x)
print(X)
```



Example of signal de-noising by FT for original noised signal (a), frequency signal by FT (b), reconstructed signal after intercepting frequency less than 10 Hz (c), reconstructed signal after intercepting frequency less than 20 Hz (d).

➤ Smoothing □ □ □ □

$$\widehat{f}(\xi) \cdot e^{i2\pi\xi x} = Ae^{i\theta} \cdot e^{i2\pi\xi x} = \underbrace{Ae^{i(2\pi\xi x + \theta)}}_{\text{polar coordinate form}} = \underbrace{A \cos(2\pi\xi x + \theta) + iA \sin(2\pi\xi x + \theta)}_{\text{rectangular coordinate form}}.$$

$$[W_\psi f](a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} \overline{\psi\left(\frac{x-b}{a}\right)} f(x) dx$$

Limitation of FTT and WT

Compared with the basic functions used in **FT (only trigonometric functions)**, the wavelet functions used in WT are not unique, that is, $f(t)$ has diversity, the same problem is analyzed with different wavelet functions sometimes the results are very different. Therefore, the selection of wavelet function is a difficulty in the practical application of WT. At present, the best wavelet function is usually selected by comparing the results of experience or continuous trial.

Software and codes



➤ Smoothing □ □ □ □

In the numerous wavelet basis function family, some wavelet functions have been proved effective in practice. The most commonly used in spectral analysis mainly include Haar, Daubechies (dbN), Coiflet, and Symlets.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pywt
from scipy.fft import fft, ifft

In [2]: # 生成含有噪声的信号
t = np.linspace(0, 1, 400)
x = np.sin(2*np.pi*10*t) + np.sin(2*np.pi*20*t) + 0.5*np.random.randn(len(t))

In [3]: # 进行小波变换
wavelet = 'db4'
coeffs = pywt.wavedec(x, wavelet)

In [4]: # 设定阈值, 进行去噪
threshold = 1.0
coeffs_thresholded = [pywt.threshold(c, threshold, mode='soft') for c in coeffs]

In [5]: # 重构信号
cleaned_signal_wavelet = pywt.waverec(coeffs_thresholded, wavelet)
```

To obtain an analytic expression:

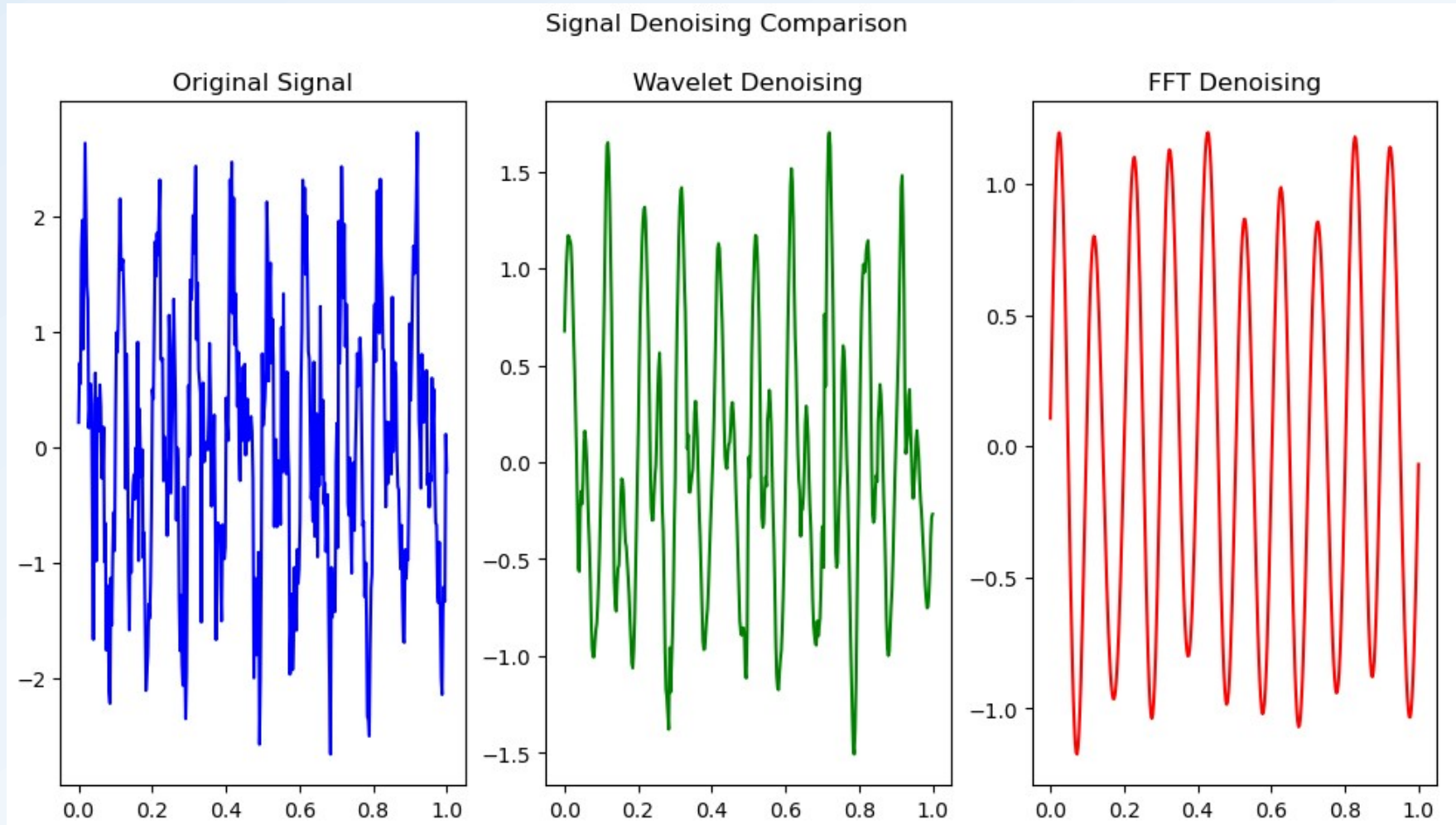


Mallat proposed multi-resolution signal decomposition (MRSD) or Pyramid is often used in numerical calculation, which is also known as **Mallat algorithm**.

Software and codes



➤ Smoothing □ □ □ □



➤ MLR, PCA, PCR, PLS

MLR ()

In practice, multiple regression methods are used in many cases to better describe the relationship between variables. Actually, the method of dealing with multivariate is basically the same as that of single variate, except that the calculation of **multiple linear regression (MLR)** is much larger, and computers are generally involved for processing.

Suppose there is a relationship between the dependent variable y and the independent variables $x_1, x_2, \dots,$

$$x_m: y = b_0 + b_1x_1 + \dots + b_mx_m$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1m} \\ 1 & x_{21} & \dots & x_{2m} \\ \dots & \dots & \dots & \dots \\ 1 & x_{n1} & \dots & x_{nm} \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_0 \\ \dots \\ b_m \end{bmatrix} \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \epsilon_1 \\ \dots \\ \epsilon_n \end{bmatrix}$$

➤ MLR, PCA, PCR, PLS

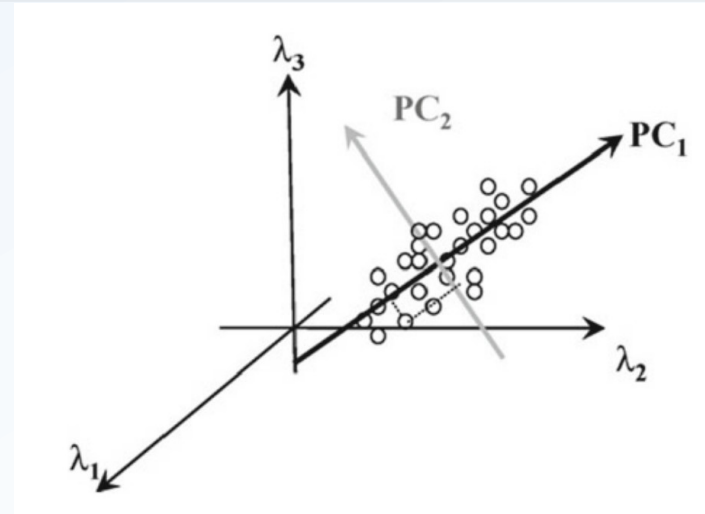
PCA ()

Principal component analysis (PCA) plays an important role in chemometrics. In fact, PCA is a very traditional technique of multivariate statistical analysis, first proposed by Hotelling in 1933.

The key purpose of PCA is to **reduce the dimension of data** and transform the original variables, so that a few new variables are linear combinations of the original variables. At the same time, these variables should express the data characteristics of the original variables as much as possible without losing of information. PCA transforms the data into a new coordinate system (for a spectra matrix with $X(n \times m)$):

$$\mathbf{X} = \mathbf{t}_1 \mathbf{p}_1^T + \mathbf{t}_2 \mathbf{p}_2^T + \mathbf{t}_3 \mathbf{p}_3^T + \dots + \mathbf{t}_n \mathbf{p}_m^T$$

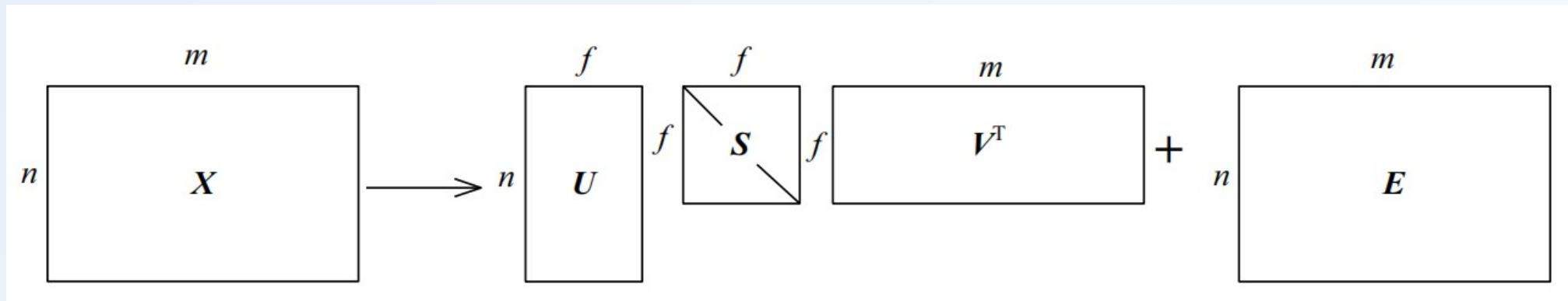
where $\mathbf{t}$ is called the score vector and $\mathbf{P}$ is called the loading vector, or principal component (PC). It can also be written in the following matrix form: $\mathbf{X} = \mathbf{T} \mathbf{P}^T$, where $\mathbf{T} = [\mathbf{t}_1 \mathbf{t}_2 \dots \mathbf{t}_n]$ is called the score matrix and $\mathbf{P} = [\mathbf{p}_1 \mathbf{p}_2 \dots \mathbf{p}_m]$ is called the loading matrix().



➤ MLR, PCA, PCR, PLS

It can be proved that the PCA of $\mathbf{X}$ is actually equivalent to the eigenvector analysis of covariance matrix $\mathbf{X}_T\mathbf{X}$ of $\mathbf{X}$. The loading vectors of matrix $\mathbf{X}$ are actually the eigenvectors of matrix $\mathbf{X}_T\mathbf{X}$. PCA and singular value decomposition (SVD,) are closely related to each other. The SVD can decompose the real number matrix of any order into the product of three matrices.

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$



where $\mathbf{S}$ is a diagonal matrix and collects the singular values of matrix $\mathbf{X}$. In fact, it is the square root of the eigenvalues of covariance matrix $\mathbf{X}_T\mathbf{X}$. $\mathbf{U}$ and $\mathbf{V}_T$ are standard column orthogonality and standard orthogonal matrix, respectively. Column and row eigenvectors corresponding to these eigenvalues are collected. In fact, the product of matrix $\mathbf{U}$ and matrix $\mathbf{S}$ is equal to the score matrix $\mathbf{T}$ in the PCA and matrix $\mathbf{V}$ is equal to the loading matrix $\mathbf{P}$.

➤ MLR, PCA, PCR, PLS

Sklearn:

```
In [1]: import numpy as np
        from sklearn.decomposition import PCA
```

```
In [2]: # 输入待降维数据 (5 * 6) 矩阵, 6个维度, 5个样本值
        A = np.array([[84, 65, 61, 72, 79, 81], [64, 77, 77, 76, 55, 70], [65, 67, 63, 49, 57, 67], [74, 80, 69, 75, 63, 74], [84, 74, 70, 80, 74, 82]])
        print(A)

        [[84 65 61 72 79 81]
         [64 77 77 76 55 70]
         [65 67 63 49 57 67]
         [74 80 69 75 63 74]
         [84 74 70 80 74 82]]
```

```
In [3]: # 直接使用PCA进行降维
        pca = PCA(n_components=2) # 降到 2 维
        pca.fit(A)
```

```
Out[3]: PCA(n_components=2)
```

```
In [4]: pca.transform(A)
```

```
Out[4]: array([[ -16.14860528, -12.48396235],
               [ 10.61676743,  15.67317428],
               [ 23.40212697, -13.607117  ],
               [-0.43966353,   7.77054621],
               [-17.43062559,   2.64735885]])
```

```
In [5]: pca.explained_variance_ratio_ # 降维后的各主成分的方差值占总方差值的比例, 即方差贡献率
```

```
Out[5]: array([0.63506778, 0.339022  ])
```

```
In [7]: pca.explained_variance_ # 降维后的各主成分的方差值
```

```
Out[7]: array([306.29319053, 163.51030959])
```

➤ MLR, PCA, PCR, PLS

In practice, the nonlinear iterative partial least squares (NIPALS,) algorithm proposed by H Wold in 1966 is often used to calculate PCA. NIPALS algorithm is more suitable for microcomputer calculation, because it is an iterative algorithm with simple calculation steps and fast speed. In the calculation, PCs with large variance or large eigenvalue are given first, rather than all factors are calculated at once. The process of NIPALS algorithm is →:

Score matrix **T** can be used as characteristic variable for quantitative analysis, such as input variable of MLR, namely, principal component regression (PCR), input variable of artificial neural network (ANN), support vector regression (SVR), etc. The score matrix **T** is also often used in qualitative analysis, such as the characteristic variable to calculate the Mahalanobis distance among samples to judge the outlier samples.

(1) Take a vector **x** in **X** as the starting value of **t**: **t** = **x**.

(2) Calculate **p^T**

$$\mathbf{p}^T = \mathbf{t}^T \mathbf{X} / \mathbf{t}^T \mathbf{t} \quad (6.6)$$

(3) Normalize **p^T**

$$\mathbf{p}^T = \mathbf{p}^T / \|\mathbf{p}\| \quad (6.7)$$

(4) Calculate **t**

$$\mathbf{t} = \mathbf{X} \mathbf{p} / \mathbf{p}^T \mathbf{p} \quad (6.8)$$

(5) Compare the new and the old **t**, and see if the convergence condition is satisfied. If the convergence condition is met, proceed to step (6), otherwise jump back to step (2).

(6) If the required PCs have been completed, the calculation will be stopped. Otherwise, calculate the residual matrix **E**

$$\mathbf{E} = \mathbf{X} - \mathbf{t} \mathbf{p}^T \quad (6.9)$$

(7) Replace **X** with **E**, go back to step (1), and find the next PC.

➤ MLR, PCA, PCR, PLS

The first f score vectors obtained by the **principal component analysis** of the spectrum matrix $\mathbf{X}$ are used to form a matrix $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_f]$, instead of absorbance variables. for MLR regression, then a principal component regression (PCR,) model is obtained as $\mathbf{y} = \mathbf{T}\mathbf{b} + \mathbf{E}$. The least square solution of regression coefficient $\mathbf{b}$ is $\mathbf{B} = (\mathbf{T}_T \mathbf{T})^{-1} \mathbf{T}_T \mathbf{Y}$.

Note that: PCR effectively overcomes the problem of unstable calculation results caused by MLR due to severe collinearity (pathological matrix,) between input variables. On the premise of maximizing the useful information in the spectra, it also suppresses the influence of measurement noise on the model by ignoring those secondary principal components and further improves the predictability of models. This method can be applied to the complex systems, and the target components can be predicted more accurately without knowing the presence of specific interfering components. **More importantly, the principle component is not necessarily relevant to the target components.**

Software and codes

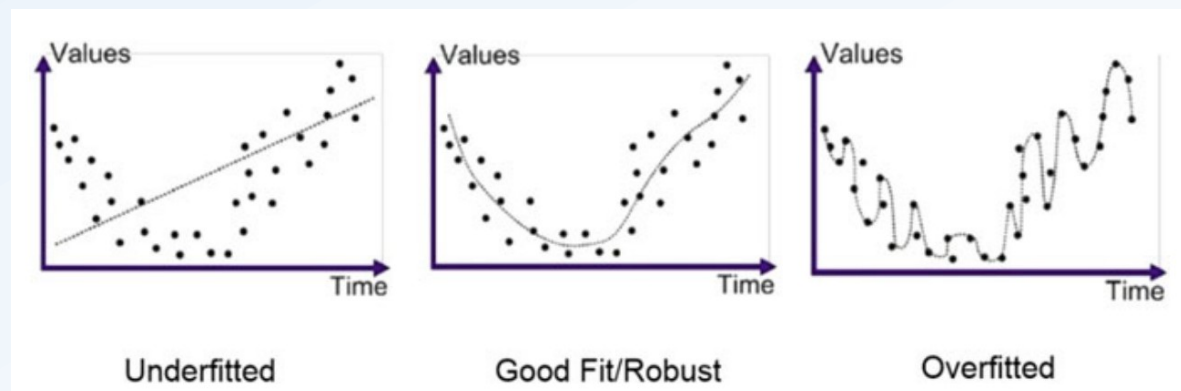


➤ MLR, PCA, PCR, PLS

In principal component regression (including the PLS introduced in the next section), it is particularly important to determine the optimal number of principal components (PCs) (also known as principal factors) participating in the regression.

If too few PCs were selected (too few features), certain useful information of the original spectra would be lost, as well, the regression fitting would be insufficient, which is called under-fitting (). Performance of under-fitting is that the obtained model has poor predictability on the calibration set samples, and the potential rules within the spectral data have not been fully learned.

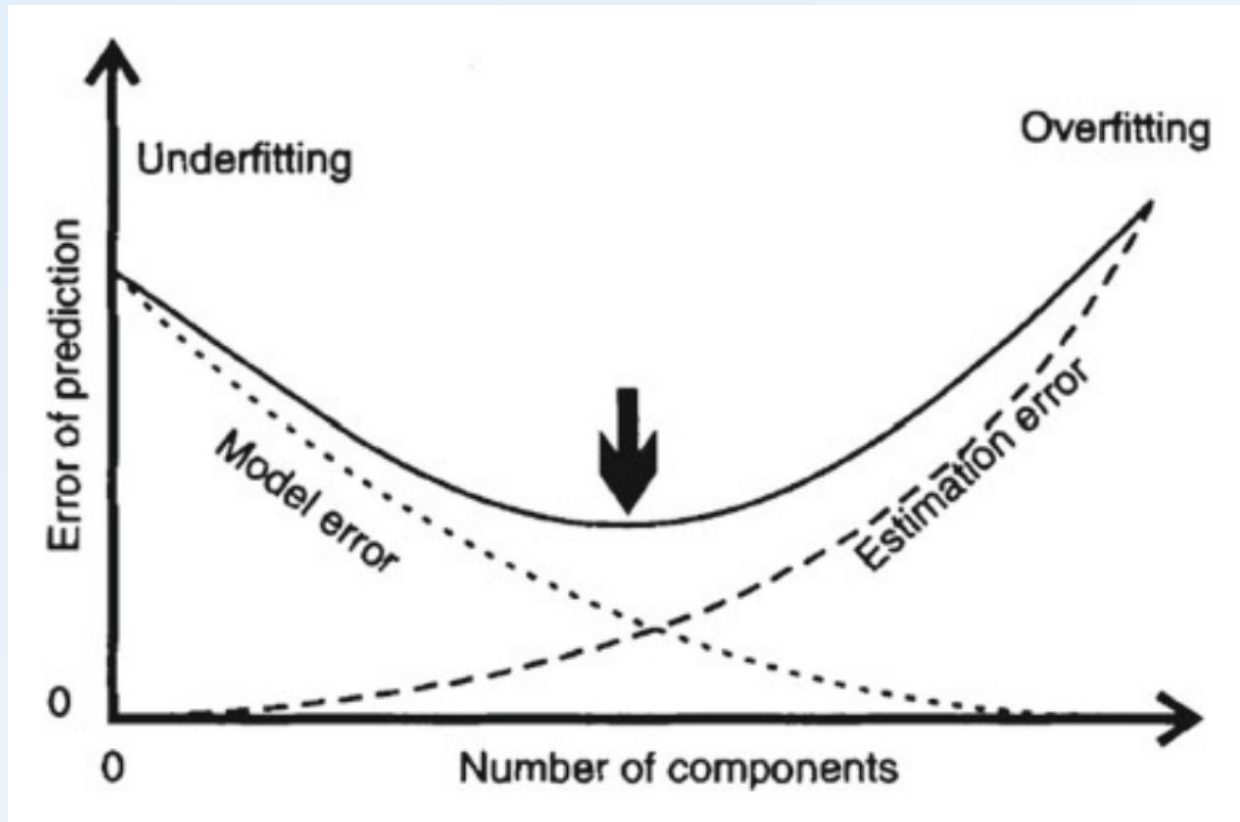
If too many PCs were selected, the measurement noise would be included too much, and the prediction error of model would increase significantly, which is called over-fitting (). **Thus, a reasonable determination of the number of PCs participating in the establishment of the model is one of the effective methods to make full use of spectral information and filter out noise.**



Software and codes



➤ MLR, PCA, PCR, PLS



For PCA: Mainly based on eigenvalues (f)

$$IND = \sqrt{\frac{\sum_{i=f+1}^{\min(n,m)} \lambda_i}{\max(n,m)[\min(n,m) - f]^5}}$$

For PCR:

Leave-one-out cross validation
Multi-fold cross validation
Monte Carlo method
Sum of ranking differences

➤ MLR, PCA, PCR , PLS

In PCR, only the spectral array $\mathbf{X}$ is decomposed to eliminate noise information. Similarly, the concentration matrix $\mathbf{Y}$ also contains useless information, which should be treated in the same way, and the influence of the concentration matrix $\mathbf{Y}$ should be considered when decomposing the spectral matrix $\mathbf{X}$. Partial least squares (PLS,) is a multiple factor regression method based on the above ideas.

First step

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}^T + \mathbf{E}_Y = \sum_{k=1}^f \mathbf{u}_k \mathbf{q}_k^T + \mathbf{E}_Y$$
$$\mathbf{X} = \mathbf{T}\mathbf{P}^T + \mathbf{E}_X = \sum_{k=1}^f \mathbf{t}_k \mathbf{p}_k^T + \mathbf{E}_X$$

where $\mathbf{t}_k(n \times 1)$ is the score of the k -th PCs of the absorbance matrix $\mathbf{X}$; $\mathbf{p}_k(1 \times m)$ is the loading of the k -th PCs of the absorbance matrix; $\mathbf{u}_k(n \times 1)$ is the score of the k -th PCs of the concentration matrix $\mathbf{Y}$; $\mathbf{q}_k(1 \times m)$ is the loading of the k -th PCs of the concentration matrix $\mathbf{Y}$; f is the number of PCs. Thus, $\mathbf{T}$ and $\mathbf{U}$ are the score matrices of $\mathbf{X}$ and $\mathbf{Y}$, $\mathbf{P}$ and $\mathbf{Q}$ are the loading matrices of $\mathbf{X}$ and $\mathbf{Y}$, $\mathbf{E}_x$ and $\mathbf{E}_y$ are the PLS fitting residual matrix of $\mathbf{X}$ and $\mathbf{Y}$, respectively.

➤ MLR, PCA, PCR , PLS

The second step of PLS is to conduct linear regression of T and U .

Second step

$$\begin{aligned} U &= TB \\ B &= (T^T T)^{-1} T^T Y \end{aligned}$$

When predicting, first the score $T_{(\text{unknown})}$ of the unknown sample spectral matrix $X_{(\text{unknown})}$ needs to be solved according to P , and then the predicted concentration value is obtained from the following equation: $Y_{(\text{unknown})} = T_{(\text{unknown})} BQ$.

In the actual PLS calculation, PLS combines matrix decomposition and regression into one step, in other words, the decomposition of X and Y matrices are performed **at the same time**. Before calculating each new PC, the score T of X is exchanged with the score U of Y , so that the PC of X is directly related to Y . Thus, when PLS calculates the PCs, it needs to consider the calculated variance of PCs to be as large as possible, while also making PCs and concentration related to the greatest extent. The **maximization of variance** () is to extract as much useful information as possible, and the maximum correlation with the concentration is to make the best use of the linear relationship between spectral variables and concentration. So it covers the shortcomings of PCR only decomposing X .

Software and codes



➤ MLR, PCA, PCR, PLS

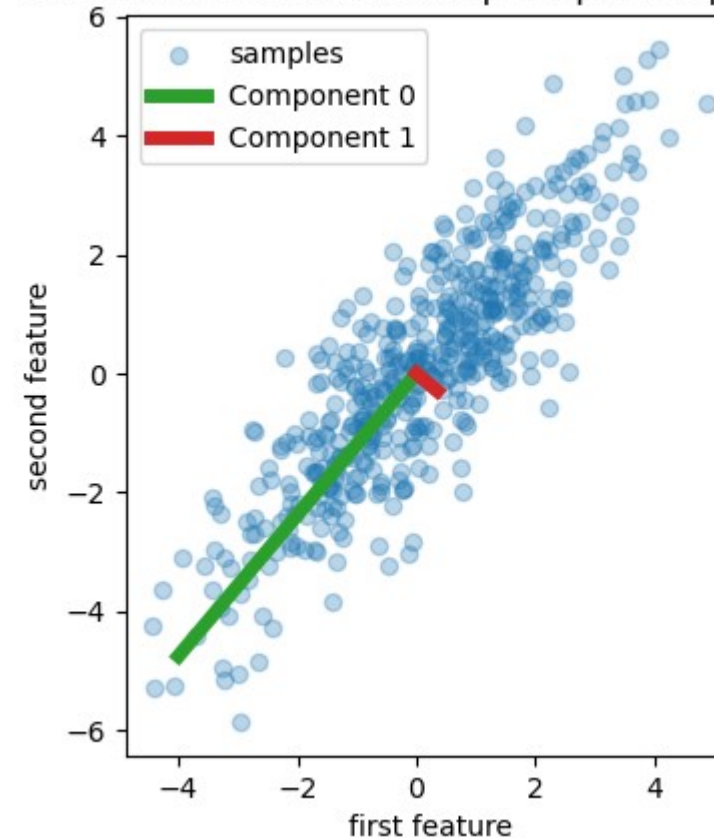
```
'''Example from : https://scikit-learn.org/stable/auto_examples/cross_decomposition/plot_pcr_vs_pls.html'''
import matplotlib.pyplot as plt
import numpy as np

from sklearn.decomposition import PCA

rng = np.random.RandomState(0)
n_samples = 500
cov = [[3, 3], [3, 4]]
X = rng.multivariate_normal(mean=[0, 0], cov=cov, size=n_samples)
pca = PCA(n_components=2).fit(X)

plt.scatter(X[:, 0], X[:, 1], alpha=0.3, label="samples")
for i, (comp, var) in enumerate(zip(pca.components_, pca.explained_variance_)):
    comp = comp * var # scale component by its variance explanation power
    plt.plot(
        [0, comp[0]],
        [0, comp[1]],
        label=f"Component {i}",
        linewidth=5,
        color=f"C{i + 2}",
    )
plt.gca().set(
    aspect="equal",
    title="2-dimensional dataset with principal components",
    xlabel="first feature",
    ylabel="second feature",
)
plt.legend()
plt.show()
```

2-dimensional dataset with principal components



Software and codes



➤ MLR, PCA, PCR, PLS

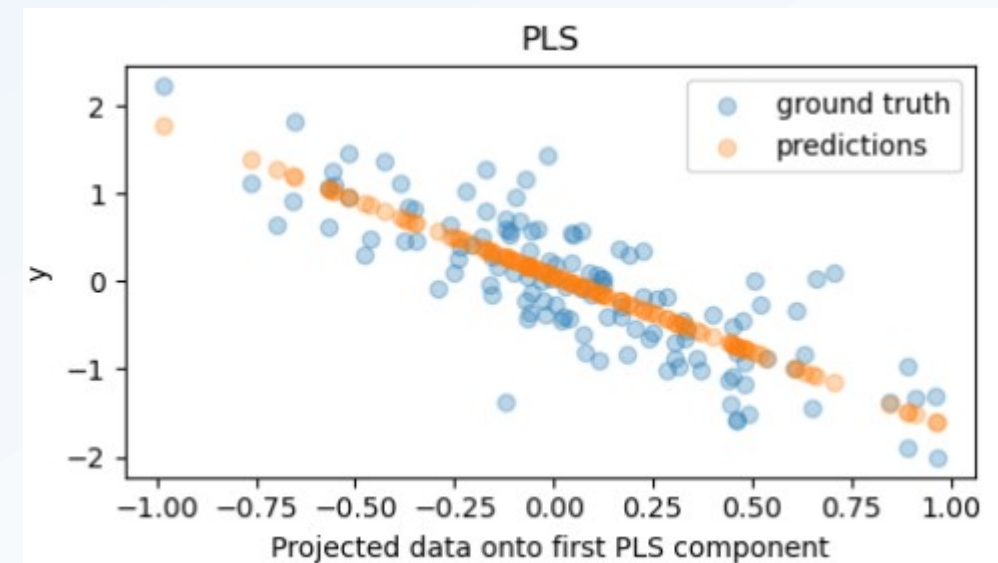
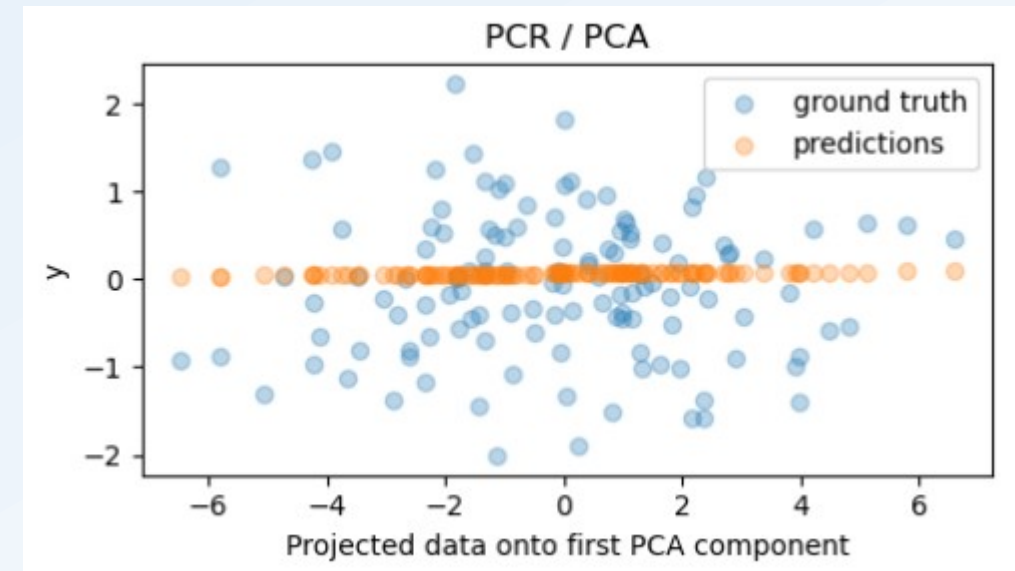
```
from sklearn.cross_decomposition import PLSRegression
from sklearn.decomposition import PCA
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=rng)

# PCR
pcr = make_pipeline(StandardScaler(), PCA(n_components=1), LinearRegression())
pcr.fit(X_train, y_train)
pca = pcr.named_steps["pca"] # retrieve the PCA step of the pipeline

# PLS
pls = PLSRegression(n_components=1)
pls.fit(X_train, y_train)

fig, axes = plt.subplots(1, 2, figsize=(10, 3))
axes[0].scatter(pca.transform(X_test), y_test, alpha=0.3, label="ground truth")
axes[0].scatter(
    pca.transform(X_test), pcr.predict(X_test), alpha=0.3, label="predictions"
)
axes[0].set(
    xlabel="Projected data onto first PCA component", ylabel="y", title="PCR / PCA"
)
axes[0].legend()
axes[1].scatter(pls.transform(X_test), y_test, alpha=0.3, label="ground truth")
axes[1].scatter(
    pls.transform(X_test), pls.predict(X_test), alpha=0.3, label="predictions"
)
axes[1].set(xlabel="Projected data onto first PLS component", ylabel="y", title="PLS")
axes[1].legend()
plt.tight_layout()
plt.show()
```

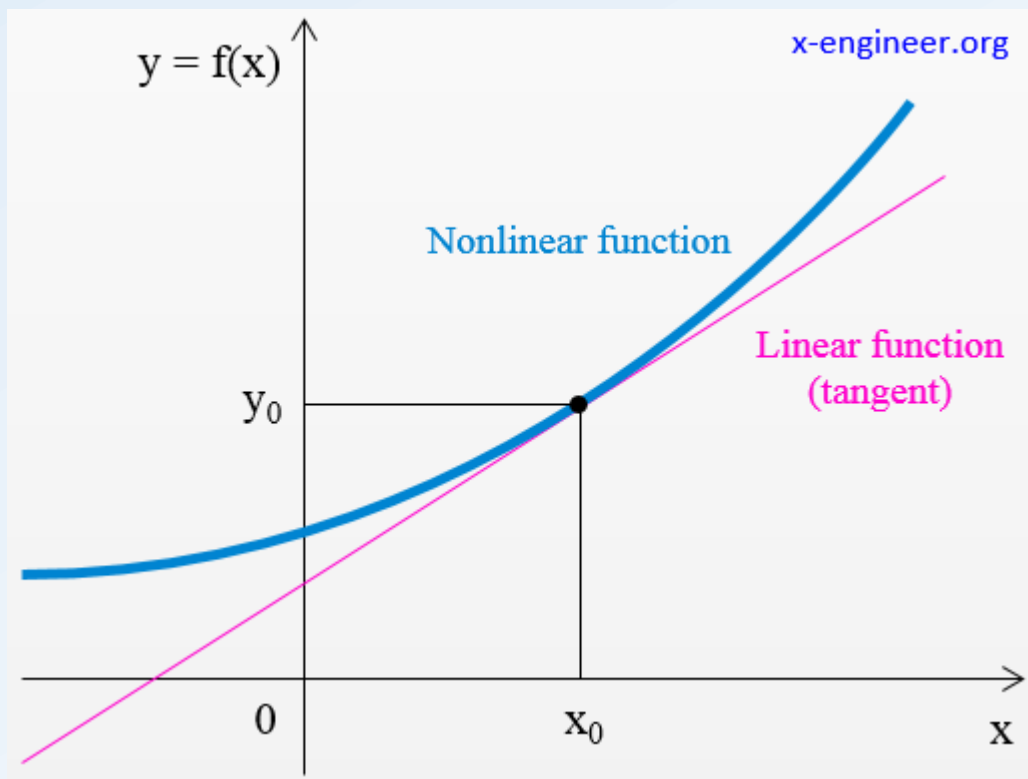


➤ MLR, PCA, PCR , PLS

From the above introduction, it can be concluded that MLR, PCR, and PLS are actually connected and coherent, and there is a gradually developed course of the linear multivariate calibration. Overcoming the weaknesses of MLR sub-rank inversion and insufficient use of spectral information, PCR uses PCA to decompose the spectral array $\mathbf{X}$, and performs MLR regression by the score vector, significantly enhancing the model prediction. PLS decomposes the spectra $\mathbf{X}$ and concentration $\mathbf{Y}$ simultaneously, strengthens the corresponding calculation relationship between the two matrices, assuring the best calibration model established. Up to now, PLSR is a perfect combination of MLR, canonical correlation analysis (CCA), and PCA. This is why PLS is most widely used in spectral multivariate calibration analysis.

Back to the question in the beginning:

Why we need to introduce AI in to chemometrics ?



Nonlinear



Big Data